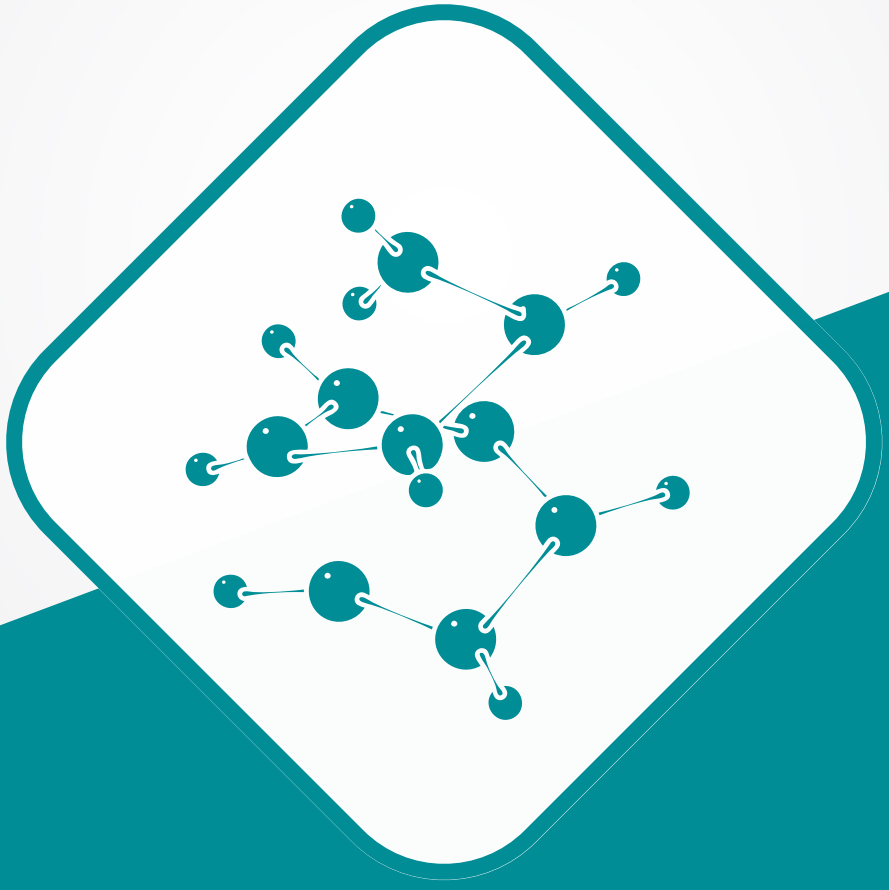
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Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Biomolecules

English Medium

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BIOMOLECULES (AMINO ACIDS AND PROTEINS)

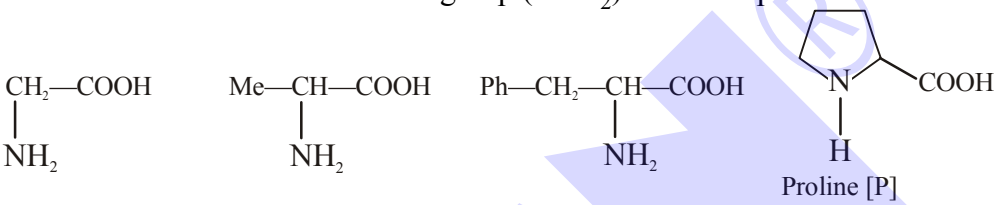
Proteins: The name protein is taken from the Greek word "proteios", which means "first". Of all chemical compounds, proteins must almost certainly be ranked first, for they are the substance of life.

Proteins make up a large part of the animal body, they hold it together and they run it. They are found in all living cells.

Chemically, proteins are high polymers. They are polyamides and the monomers from which they are derived are the α - amino carboxylic acids. A single protein molecule contains hundreds or even thousands of amino acid units. These units can be of twenty-odd different kinds. The number of different combinations, i.e., the number of different protein molecules that are possible, is almost infinite.

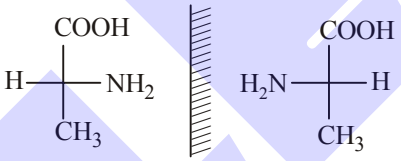
Bifunctional compounds having an acidic corboxylic group & a basic amino group are known as amino acid.

There are 20 amino acids commonly found is proteins and are standard amino acids. All are α amino acids. Most of them have 1° amino group.($-NH_2$). However proline is a 2° amino.



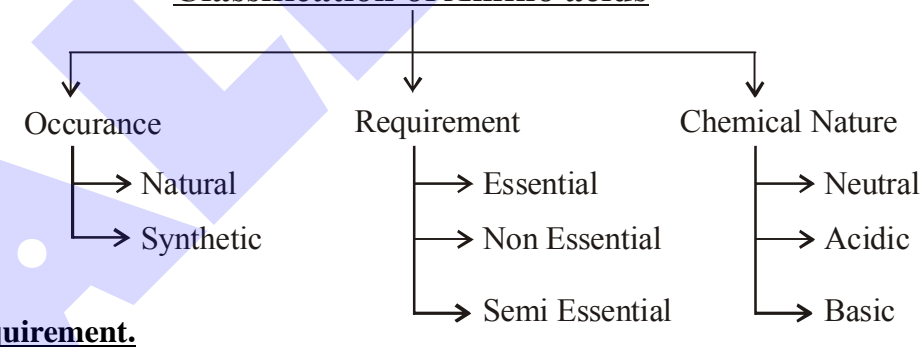
(Gly) Glycine [G] Alanine [A] Phenyl alanine [F] Proline [P] (α-imino acid)

All amino acids are chiral molecules with atleast one chiral carbon (except glycine, $H_3N^{\oplus}CH_2COO^{-}$). Except Glycine all other amino acids are optically active & can be assigned D & L configuration.



D-Alanine L-Alanine

Classification of Amino acids



Based on requirement.

1. Essential amino acids can not be synthesized in human body so dietary intake is required. For any human being 1 gm a day is required.
2. Semi essential amino acids can be synthesized in human body but dietary intake is required during growing stages (when more of cell division is required).
   For example : Early childhood, pregrancy and lactating mother.
3. Non essential amino acid - Body can synthesize them.

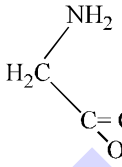
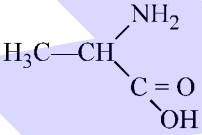
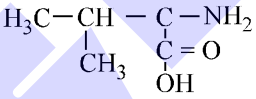
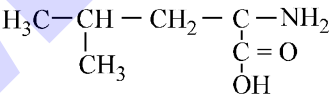
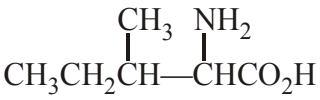
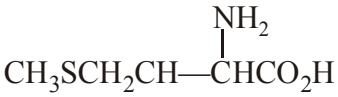
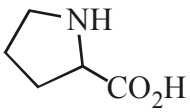
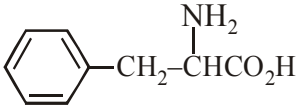
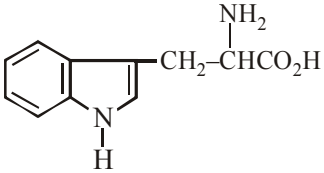
Based on chemical nature

Neutral - Amino acid having equal number of NH_2 and COOH.

Neutral amino acids are further classified as polar and nonpolar depending on whether their side chains have polar substituents (for example, asparagine with an NH_2CO group) or are completely hydrocarbon in nature (for example alanine, valine etc.).

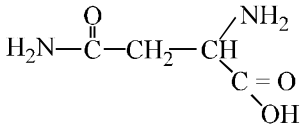
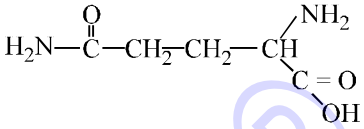
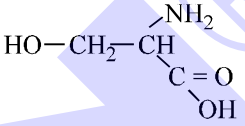
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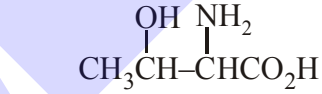
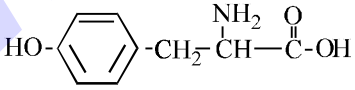
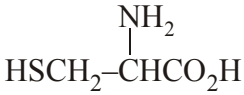
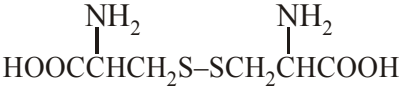
Acidic - Amino acid having more COOH than NH_2 group.
For example : Aspartic acid and glutamic acids, each with a second CO_2H in their side chain are acidic amino acids.
Basic - Amino acid having more NH_2 than COOH group.
For example : Lysine, arginine and histidine)

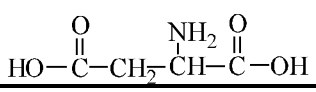
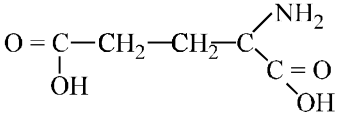
A -1. Neutral amino acids (with nonpolar side chains)			
NAME	**ABBREVIATIONS**	**STRUCTURAL FORMULAE**	**ISOELECTRIC POINT[pI]**
@Glycine**	Gly(G)		6.0
Alanine**	Ala(A)		6.0
Valine*	Val(V)		6.0
Leucine*	Leu(L)		6.0
Isoleucine*	Ile(I)		6.0
Methionine*	Met(M)		5.7
@@Proline	Pro(P)		6.3
Phenylalanine*	Phe(F)		5.5
Tryptophan*	Trp(W)		5.9

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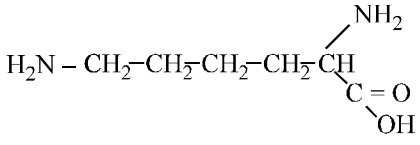
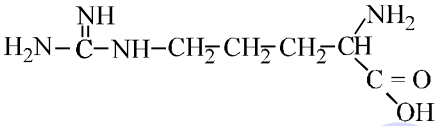
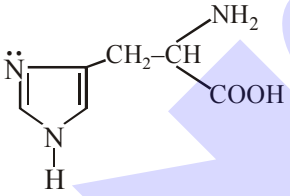
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A -2. Neutral amino acids (with polar, but nonionized side chains)			
NAME	**ABBREVIATIONS**	**STRUCTURAL FORMULAE**	**ISOELECTRIC POINT[pI]**
Asparagine**	Asn(N)		5.4
Glutamine**	Gln(Q)		5.7
Serine**	Ser(S)		5.7

A -3. Neutral amino acids (with polar, but nonionized side chains)			
NAME	**ABBREVIATIONS**	**STRUCTURAL FORMULAE**	**ISOELECTRIC POINT[pI]**
Threonine*	Thr (T)		5.6
Tyrosine**	Tyr(Y)		5.7
Cysteine	Cys (C)		5.1
‡ Cystine	Cys–Cys		

B - Acidic amino acids (side chain with carboxylic acid group)			
NAME	**ABBREVIATIONS**	**STRUCTURAL FORMULAE**	**ISOELECTRIC POINT[pI]**
Aspartic acid**	Asp(D)		2.8
Glutamic Acid	Glu(E)		3.2

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C - Basic amino acids (side chain with nitrogenous basic group)			
NAME	**ABBREVIATIONS**	**STRUCTURAL FORMULAE**	**ISOELECTRIC POINT[pI]**
Lysine*	Lys(K)		9.7
Arginine*	Arg(R)		10.8
Histidine*	His(H)		7.6

Note:

* Amino acids with an asterisk are essential amino acids.

** Amino acids with an asterisk are non essential amino acids.

† At pH = 7, Asp and Glu have a net negative charge and exist as anions. At pH = 7, Lys and Arg have a net positive charge and exist as cations. Rest of the amino acids at this pH exist in the neutral form.

‡ Structurally, in cystine, the two cysteine molecules are joined through sulfur (disulfide linkage).

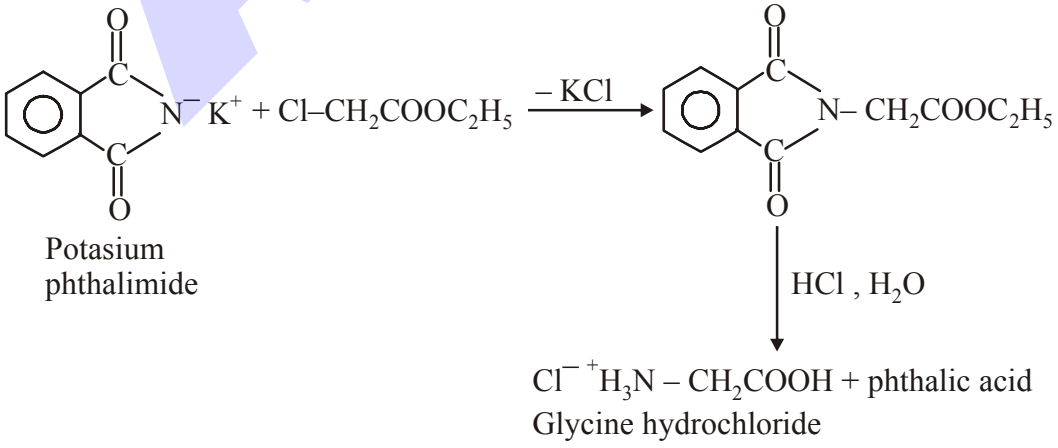
@@ Proline is an α-imino acid, all amino acids are primary amines except proline and 4-hydroxyproline, which are 2° amines.

@ Except Glycine all other amino acids are optically active.

Preparation of amino acids

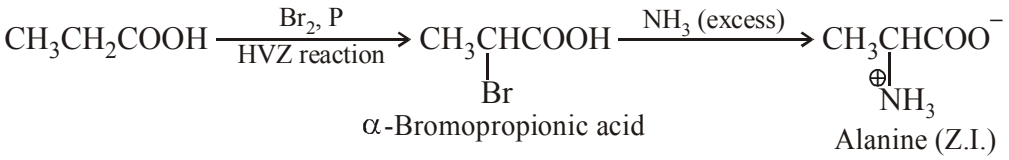
(a) **Gabriel Phthalimide synthesis**

Good yields of amino acids are generally obtained by the Gabriel phthalimide synthesis ; In this method α - halo esters are used instead of α - halo acids .

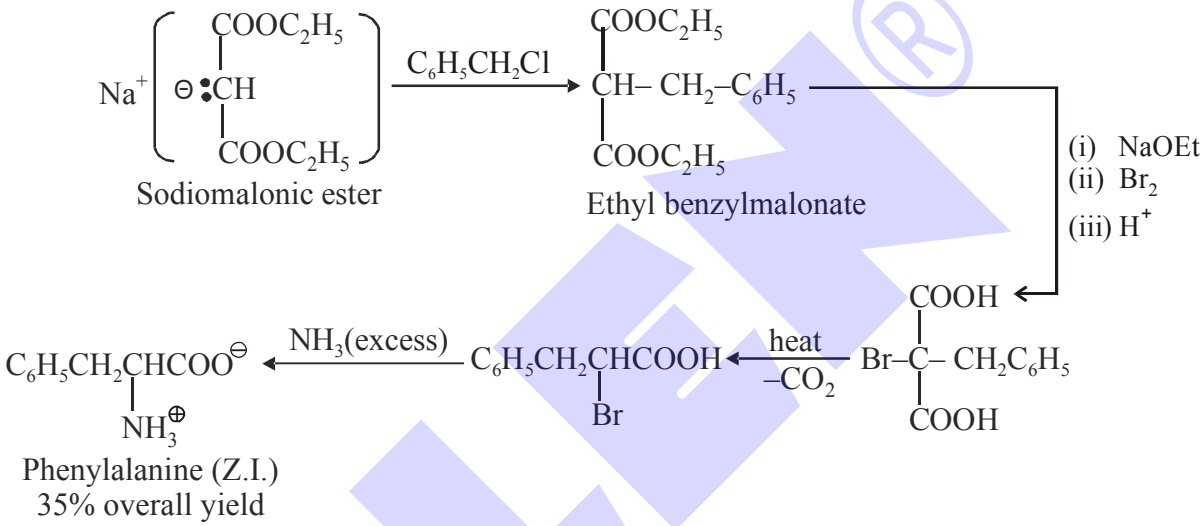


(b) **Amination of α - Halo acids**

Sometimes an α - chloro or α - bromo acid is subjected to direct ammonolysis with excess of concentrated ammonia.

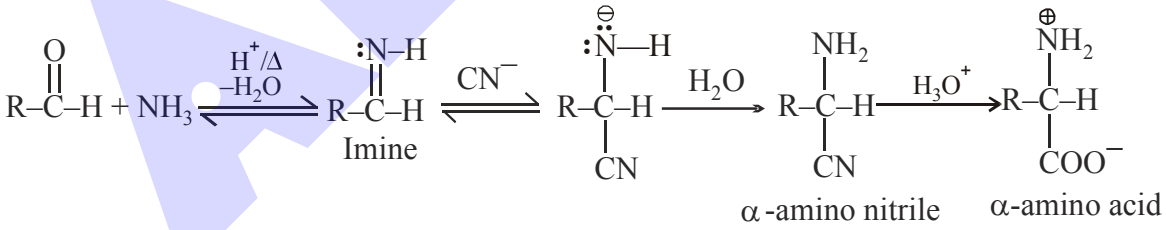


(c) **From diethyl malonate**

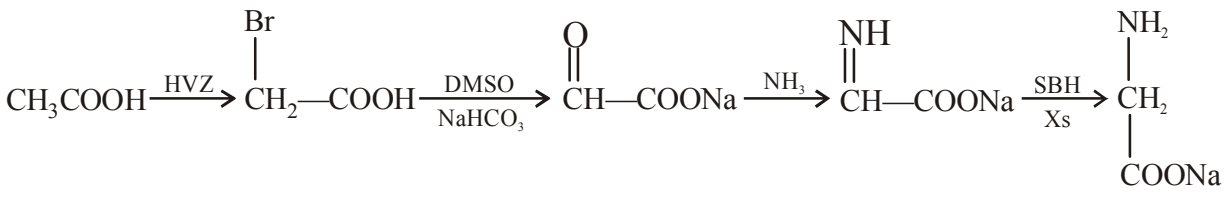


(d) **Strecker's synthesis**

Strecker's synthesis is also used for preparing α - amino acids



(e) **Using KOOP synthesis**



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Properties of Amino acids

Although the amino acids are commonly shown as containing an amino group and a carboxyl group, $H_2NCHRCOOH$, certain properties, (both physical and chemical) are not consistent with this structure

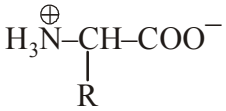
I. Physical properties

In contrast to amines and carboxylic acids, the amino acids are nonvolatile crystalline solids, which melt with decomposition at fairly high temperatures. They are insoluble in non-polar solvents like petroleum ether, benzene or ether and are appreciably soluble in water. Their aqueous solutions behave like solutions of substances of high dipole moment due to existance.

Amino acids as dipolar ions as zwitter ion

Acidity and basicity constant are ridiculously low for –COOH and $–NH_2$ groups. Glycine, for example, has $K_a = 1.6 \times 10^{-10}$ and $K_b = 2.5 \times 10^{-12}$, whereas most carboxylic acids have K_a values of about 10^{-5} and most aliphatic amines have K_b values of about 10^{-4}.

All these properties are quite consistent with a dipolar ion structure for the amino acids (I)



Amino acids : dipolar ions (Zwitter ion)

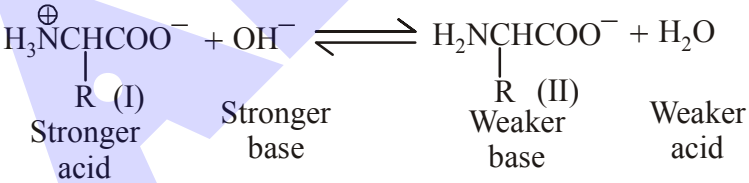
Physical properties - melting point, solubility, high dipole moment - are just what would be expected of such a salt. The acid-base properties also become understandable when it is realized that the measured K_a actually refers to the acidity of an ammonium ion, RNH_3^+,



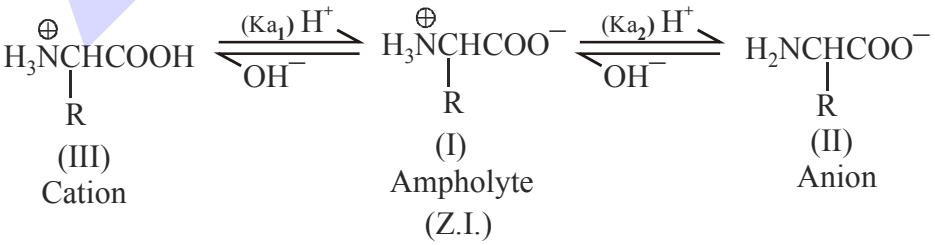
Acid

$$K_a = \frac{[H_3O^{\oplus}][H_2NCHRCOO^{\ominus}]}{[H_3\overset{\oplus}{N}CHRCOO^{\ominus}]}$$

When the solution of an amino acid is made alkaline, the dipolar ion (I) is converted into the anion(II). The stronger base, hydroxide ion, removes a proton from the ammonium ion and displaces the weaker base, the amine.



Stronger acid — Stronger base — (I) — Weaker base — (II) — Weaker acid



(III) Cation — (I) Ampholyte (Z.I.) — (II) Anion

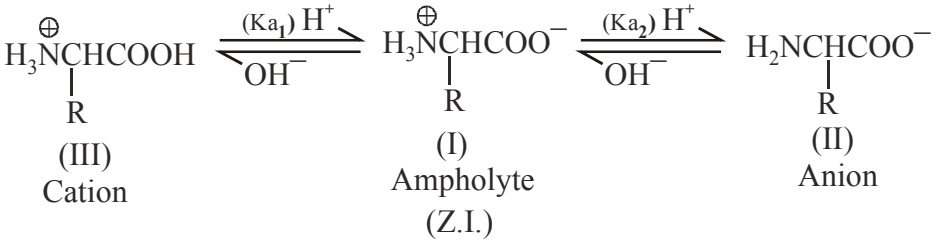
Wherever feasible, we can speed up a desired reaction by adjusting the acidity or basicity of the solution in such a way as to increase the concentration of the reactive species.

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Isoelectric point of amino acids

What happens when a solution of an amino acid is placed in an electric field depends upon the of the solution.



In quite alkaline solution, anions (II) exceed cations (III), and there is a net migration of amino acid toward the anode. In quite acidic solution, cations (III) are in excess, and there is a net migration of amino acid toward the cathode. If (II) and (III) are exactly balanced, there is no net migration ; under such conditions any one molecule exists as a positive ion and as a negative ion for exactly the same amount of time, and any small movement in the direction of one electrode is subsequently cancelled by an equal movement back towards the other electrode. **The hydrogen ion concentration of the solution in which a particular amino acid does not migrate under the influence of an electric field is called the isoelectric point (pI) of that amino acid. The isoelectric point (pI) is the pH at which the amino acid exists only as a dipolar ion with net charge zero.**

For glycine, for example, the isoelectric point is at pH 6.1.

An amino acid usually shows its lower solubility in a solution at the isoelectric point, since here there is the highest concentration of the dipolar ion. As the solution is made more alkaline or more acidic, the concentration of one of the more soluble ions, (II) or (III) increases.

$$Ka_1 = \frac{[Z.I.]\ [H^{\oplus}]}{[C^{\oplus}]} \qquad Ka_2 = \frac{[A^{-}][H^{\oplus}]}{[DI]} \quad \text{at pI } [A^{\ominus}] = [C^{\oplus}]$$

$$\frac{[Z.I.]\ [H^{\oplus}]}{Ka_1} = \frac{Ka_2[DI]}{[H^{\oplus}]} \qquad [H^{\oplus}]^2 = Ka_1 \ \& \ Ka_2$$

$$\text{on taking log pI} = \frac{P^{Ka_1} + P^{Ka_2}}{2}$$

An amino acid having –COOH group more than NH_2 group or such amino acid have pI less than 7.

An amino acid having more – NH_2 more than COOH group such amino acid have pI more than 7.

Q. Write the structure of alanine at pH 2.5, 10.5 and 6.

Electrophoresis

The movement of charged molecules (like amino acid) under the influence of an electric field is called electrophoresis. Electrophoresis separates amino acids on the basis of their pI values.

Amino acid is positively charged (moves towards cathode) if pH of the solution < pI

Amino acid is negatively charged (moves towards anode) if pH of the solution > pI

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Q. How will you separate a ternary mixture of arginine, alanine & aspartic acid?

Ans. A few drops of a solution of an amino acid mixture are applied to the middle of a piece of filter paper. When the paper is placed in a buffer solution (pH = 5) between the two electrodes and an electric field is applied then arginine & alanine with pI > pH move towards the cathode and aspartic acid with pI < pH moves towards the anode. Out of arginine & alanine, alanine will move slowly towards the cathode due to lesser positive charge.

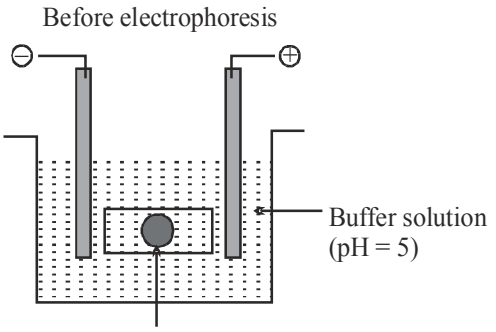


Figure (i)

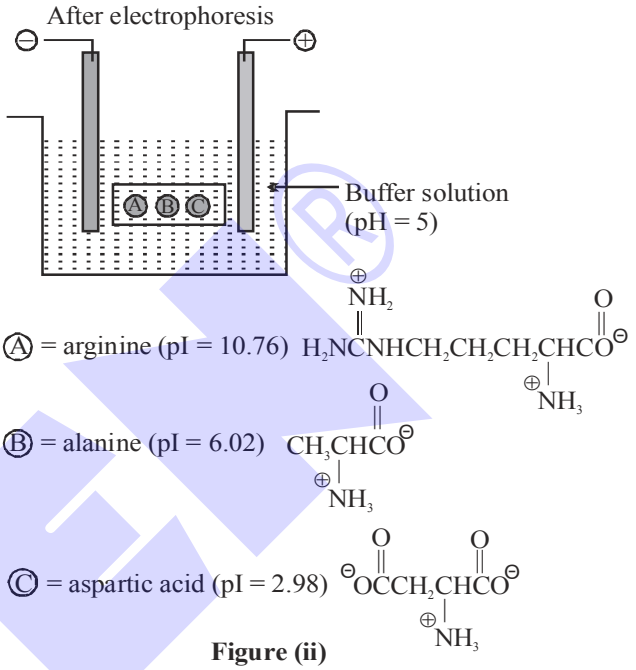
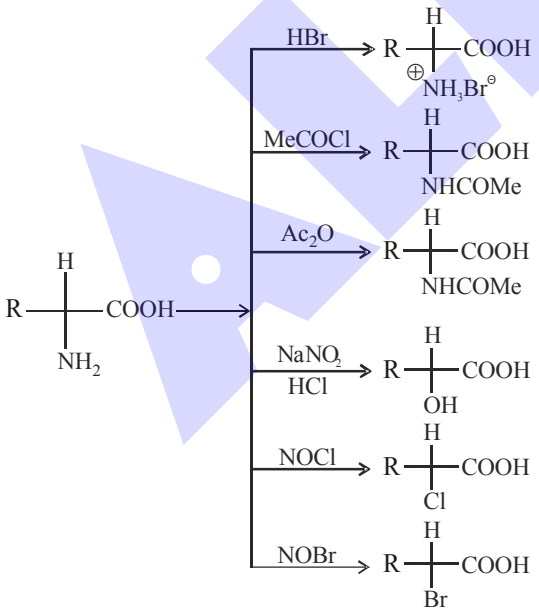
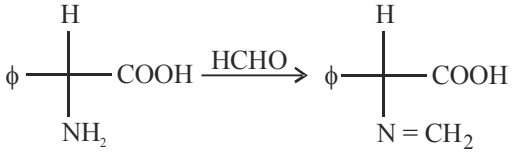


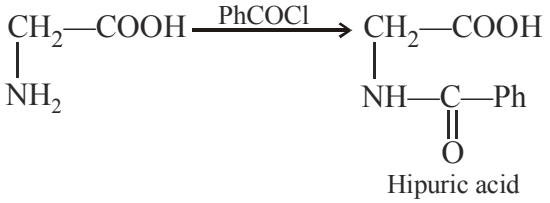
Figure (ii)

General reactions of amino acids

(1) **Reactions due to – NH_2 group**



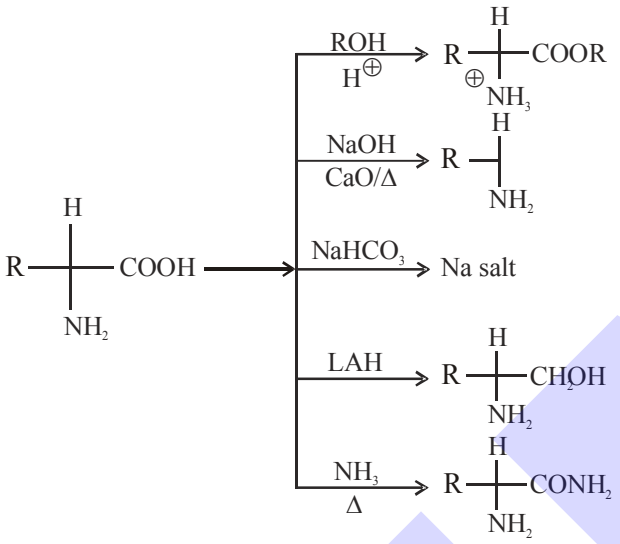




Hipuric acid

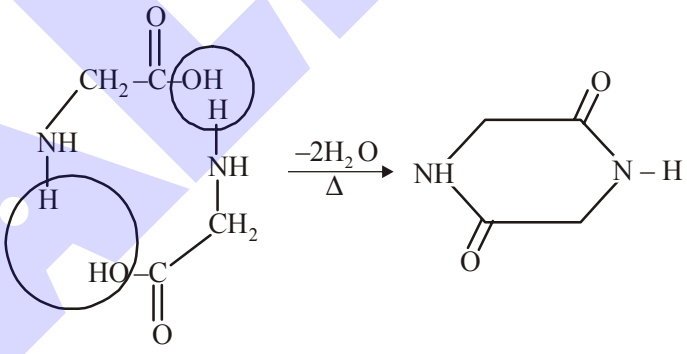
Reactions is used to block – NH_2 group during volumetric analysis in.

(2) **Reactions due to – COOH group.**

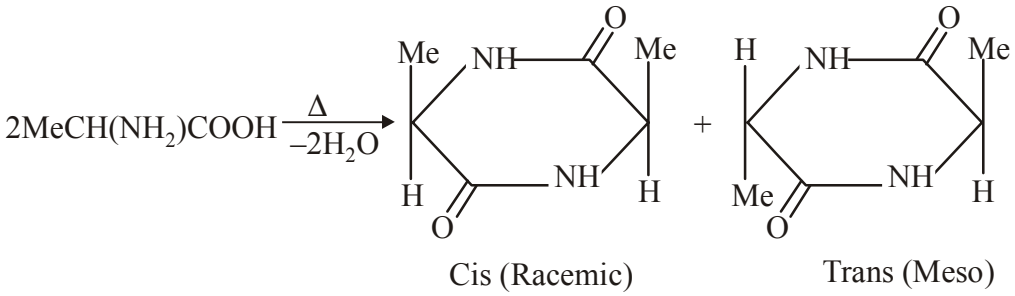


(3) **<u>Heating Effect</u>**

(i) Heating of amino acids leads to intermolecular dehydration to form cyclic diamides.



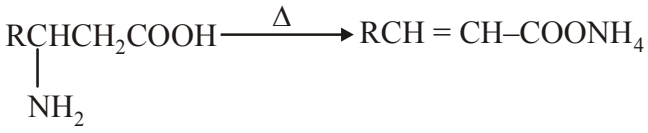
(ii) When alanine is heated, then two diastereomers are obtained. One of them (trans) is not resolvable.



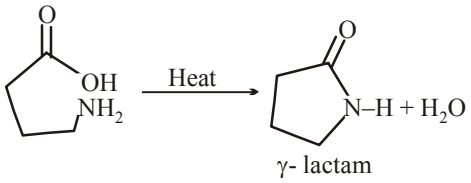
Cis (Racemic) Trans (Meso)

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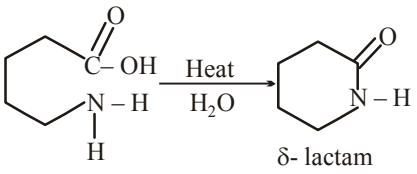
(iii) When β - amino acids are heated, α, β - unsaturated salt are formed.



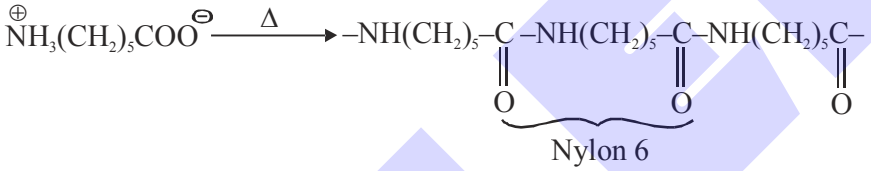
(iv) γ,δ,ε - amino acids when heated alone gives γ, δ - lactam and polymer respectively. The reason for the formation of polymer is that when ε - amino cyclises intramolecularly, it leads to large angle strain within the compound



γ- lactam



δ- lactam



Nylon 6

(4) **Peptide**

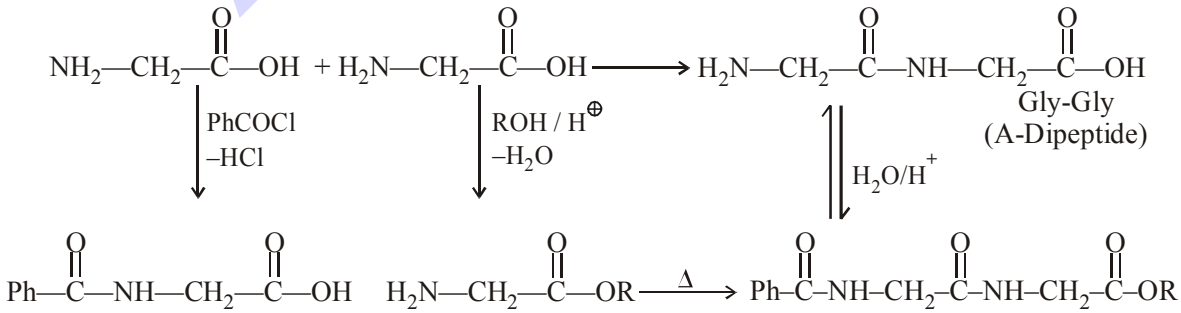
A peptide is a kind of amide formed by intermolecular reaction of the amino group of one amino acid and the carboxyl group of a second amino acid. Dipeptides are made from two amino acids, tripeptides from three amino acids, etc, which may be the same or different. If there are three to ten amino acid residues, the peptide is also called an oligopeptide.

If they give 3 to 10 amino acid they are oligopeptide

If they give 11 to 100 amino acid they are Polypeptide

For more than 100 it is Macropeptide

- Peptides can be prepared by blocking technique



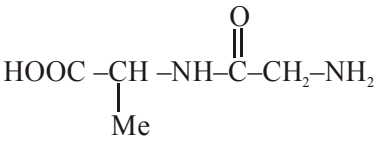
Gly-Gly
(A-Dipeptide)

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Firstly, the amino and carboxyl groups that are not to be linked in peptide bonds must be blocked to make unreactive.

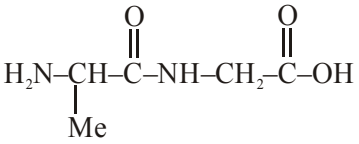
Abbreviated name of amion acid with free NH_2 is written first.



Dipeptide

Gly–Ala [Glycine Alanine]

By convention, the amino acid with the free amino group (N-terminal) is written at the left end and the one with the unreacted carboxyl group (C-terminal) at the right end.



Ala - Gly [Alanine-Glycine]

- When different amino acids are involved in pepetide formation.

Then total number of polypeptide possible = X^n

[X = type of amino acid interacting,

n = number of amino acid molecule are interacting.]

Q. Glycine can form how many Dipeptide ? [Ans. One]

Q. Glycine can form how many Tripeptide ? [Ans. One]

Q. Glycine and Ala can form how many Dipeptide ? [Ans. Four]

Q. Gly, Ala, and Phenyl Ala can form how many Dipeptide ? [Ans. Nine]

Q. Gly, Ala, can form how many Tripeptide ? [Ans. Eight]

A polypeptide with more than hundred amino acid residues, having molecular mass higher than 10,000 is called a protein. However, the distinction between a polypeptide and a protein is not very sharp. Polypeptides with fewer amino acids are likely to be called proteins they ordinarily have a well defined conformation of a protein such as insulin which contains 51 amino acids.

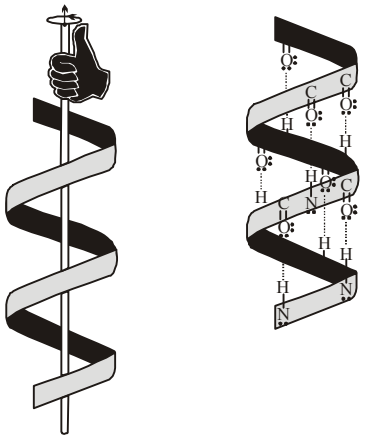
STRUCTURE OF PROTEINS

(i) **Primary structure of proteins :** Proteins may have one or more polypeptide chains. Each polypeptide in a protein has amino acids linked with each other in a specific sequence and it is this sequence of amino acids that is said to be the primary structure of the protein. Any change in this primary structure i.e., the sequence of amino acids creates a different protein.

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(ii) **Secondary structure of proteins :** The secondary structure of protein refers to the shape in which a long polypeptide chain can exist. They are found to exist in two different types of structures viz. α-helix and β-pleated sheet structure. These structures arise due to the regular folding of the backbone of the polypeptide chain **due to hydrogen bonding** between

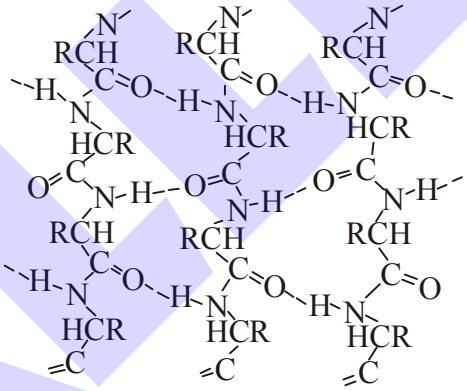
$$-\overset{\overset{O}{\|}}{C}-$$ and — NH — groups of the peptide bond.



α-Helix structure of proteins

α-Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a **right handed screw (helix)** with the — NH group of each amino acid residue hydrogen bonded to the $>C=O$ of an adjacent turn of the helix as shown in figure.

In β-structure all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together, by intermolecular hydrogen bonds. The structure resembles the pleated folds of drapery and therefore is known as β-pleated sheet.



β-Pleated sheet structure of proteins

Different types of bonding present in different structures of protein.

(i) Peptide bond : It is present in primary structures of proteins.

(ii) H-bonding : mainly between side-chain NH_2 and COOH, also involving OH's (Of serine, for example) and the N–H of tryptophan. (It is present in secondary structures of protein)

(iii) Ionic bonding : between COO^- and NH_3^+ at different sites.

(iv) Weakly hydrophobic Van der Waal's attractive forces engendered by side-chain R groups and

(v) Disulfide cross linkages between loops of the polypeptide chain.

The same kind of attractive and repulsive forces responsible for the tertiary structure operate to hold together and stabilize the subunits of the quaternary structure.

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(iii) **Tertiary structure of proteins :** The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 3° structures of proteins are hydrogen bonds, disulphide linkages, van der walls and electrostatic forces of attraction.

There are two main classes of protein tertiary structure:

(a) **Fibrous proteins**

When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed which is known as fibrous protein.

(b) **Globular proteins**

This structure results when the chains of polypeptides coil around to give a spherical shape. Which is known as globular protein.

	Fibrous	Globular
Shape	Long and narrow	Round / spherical
Purpose	Structural protein	Functional protein
Acid sequence	Repetitive amino acid sequence	Irregular amino acid sequence
Durability	Less sensitive to changes in pH, temperature, etc.	More sensitive to changes in pH, temperature, etc.
Solubility	(Generally) insoluble in water	(Generally) soluble in water
Examples	Keratin (Present in hair, wool silk) Actin and Myosin (Present in musles), Fibrin, Collagen and Elastin etc.	Albumin (egg protein), Enzyme, Haemoglobin, Insulin, Immunoglobulin (Antibody) (Ig) etc.

(iv) **Quaternary structure of proteins :** Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. The spatial arrangement of these subunits with respect to each other is known as quaternary structure. Example : Heamoglobin, Chlorophyll.

According the their biological action, they are classified as enzymes, hormones, antibodies, etc.

Protein found in living system with definite configuration and biological activity is termed as native protein. If a native protein is subjected to physical or chemical treatment, which may disrupt its higher structures (conformations) without affecting its primary structure, the protein is said to be denatured. During denaturation, the protein molecule uncoils form an ordered and specific conformation into a more random conformation leading to precipitation. Thus denaturation leads to increase in entropy and loss of biological activity of the protein. The denaturation may be reversible or reversible. Thus, the coagulation of egg white on boiling of egg protein is an example of irreversible protein denaturation. However, in certain cases it is found that if the disruptive agent is removed the protein recovers its original physical and chemical properties and biological activity the reverse of denaturation is known as renaturation.

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Test Of Amino acids and Proteins

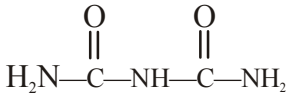
1. Biurate Test **2. Nin hydrin Test** **3. Xanthoproteic test**

4. Sakaguchi test **5. Millon's Test**

1. Biurate Test

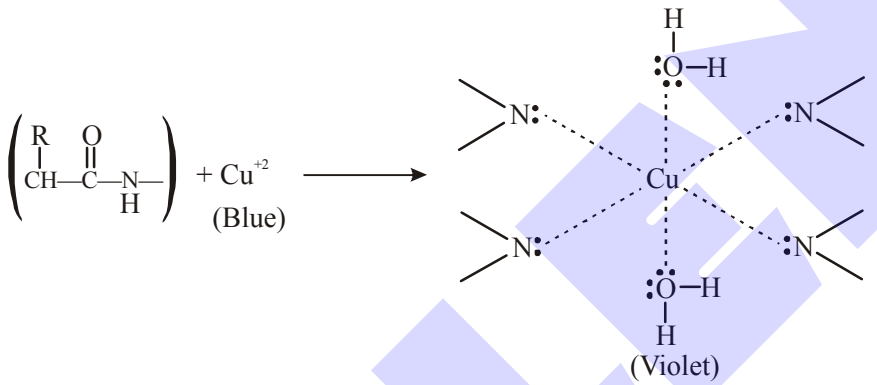
Addition of a very dilute solution of $CuSO_4$ to an alkaline solution of a protein is done. A positive test is indicated by the formation of a pink violet to purple violet color.

The name of test is derived from a specific compound, biuret, which gives a positive test with this reagent

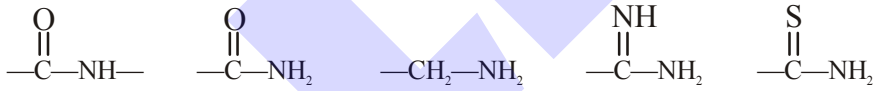


biuret

When a protein reacts with copper (II) sulfate (blue), the positive test is the formation of a violet colored complex.



The biuret test works for any compound containing two or more of the following groups.

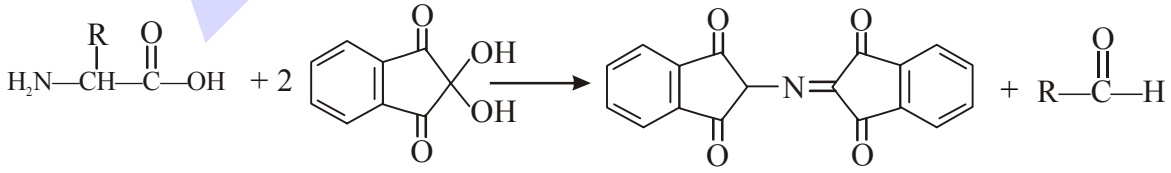


2. Nin hydrin Test

The ninhydrin test is a test for amino acids and proteins with a free $-NH_2$ group.

Amino acids are detected by ninhydrin test. All amino acids give violet - coloured product with ninhydrin (triketo hydroindene hydrate) except proline and 4 - hydroxy proline, which gives yellow colour with it.

When such an $-NH_2$ group reacts with ninhydrin, a purple-blue complex is formed.



The same violet coloured dye forms from all α - AA's with 1° amino groups because only their nitrogen is incorporated into it. The 2° amines proline and 4 - hydroxyproline give different adducts that absorb light at a different and thus have a different yellow colour.

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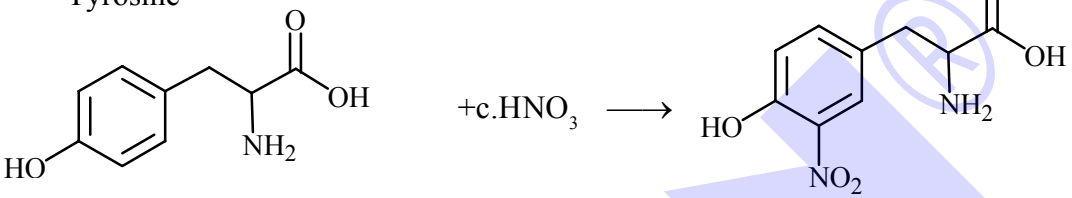
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3. Xanthoproteic test

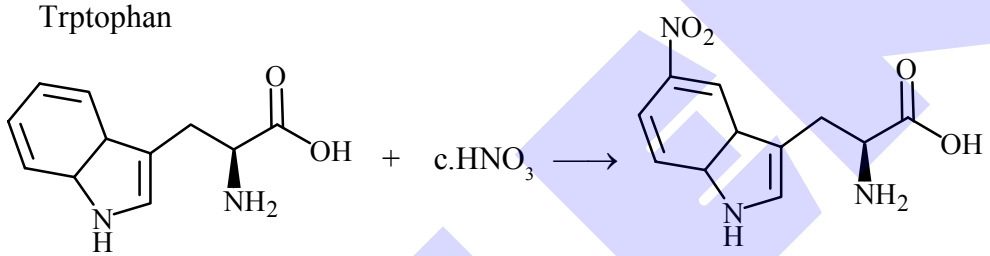
This test is used for aromatic amino acids which give positive result from other amino acids. Such as tyrosine, and tryptophan gives Xanthoproteic test, phenyl alanine does not respond with this test.

Principle : Xanthoproteic test is used to detect amino acids containing an aromatic nucleus (tyrosine, tryptophan and phenylalanine) in a protein solution which gives yellow color nitro derivatives on heating with conc. HNO_3. The aromatic benzene ring undergoes nitration to give yellow colored product. Phenylalanine gives negative or weakly positive reaction though this amino acid contains aromatic nucleus because it is difficult to nitrate under normal condition. On adding alkali to these nitro derivative salts, the color change for yellow to orange.

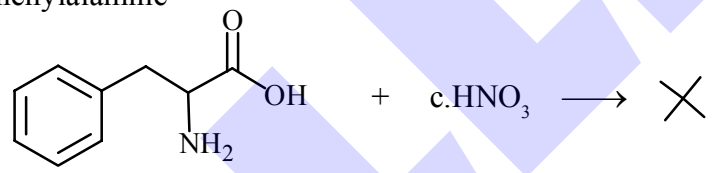
Tyrosine



Trptophan



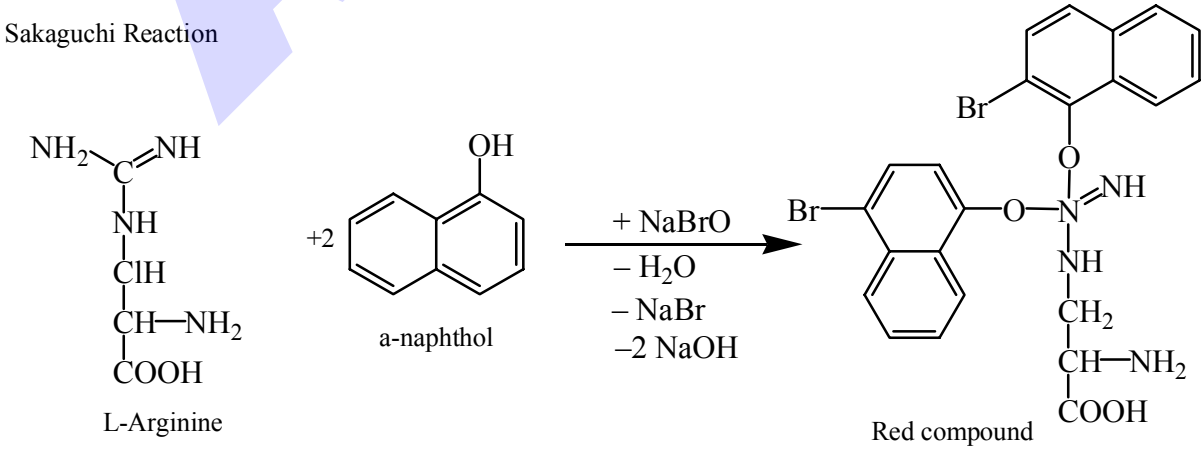
Phenylalamine



4. Sakaguchi test

The Sakaguchi test is a chemical test used for detecting the presence of arginine in proteins Sakaguchi reagent consists of α-Naphthol and a drop of sodium hypobromite. The guanidine group in arginine reacts with Sakaguchi reagent to form a red-coloured complex.

Sakaguchi Reaction



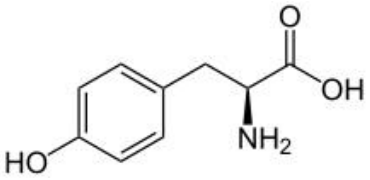
L-Arginine

Red compound

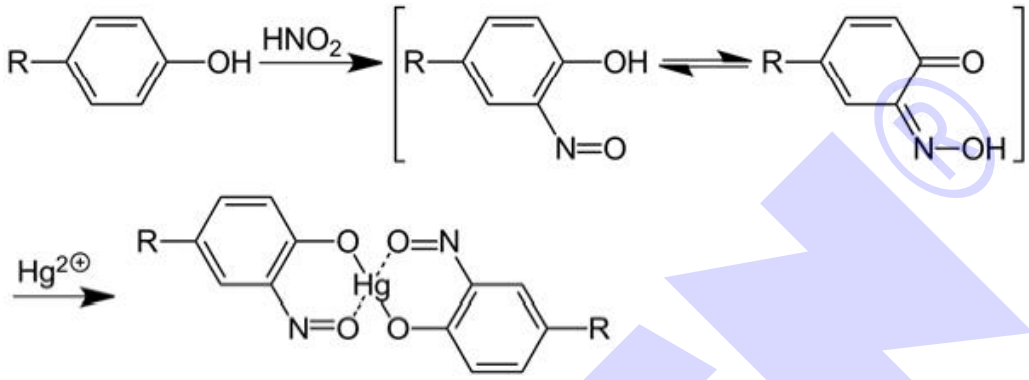
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5. Millon's Test

Millon's reagent is an analytical reagent used to detect the presence of soluble proteins. A few drops of the reagent are added to the test solution, which is then heated gently. A reddish-brown coloration or precipitate indicates the presence of tyrosine residue which occur in nearly all proteins



Tyrosine Here R = $-CH_2CH(NH_2)COOH$.



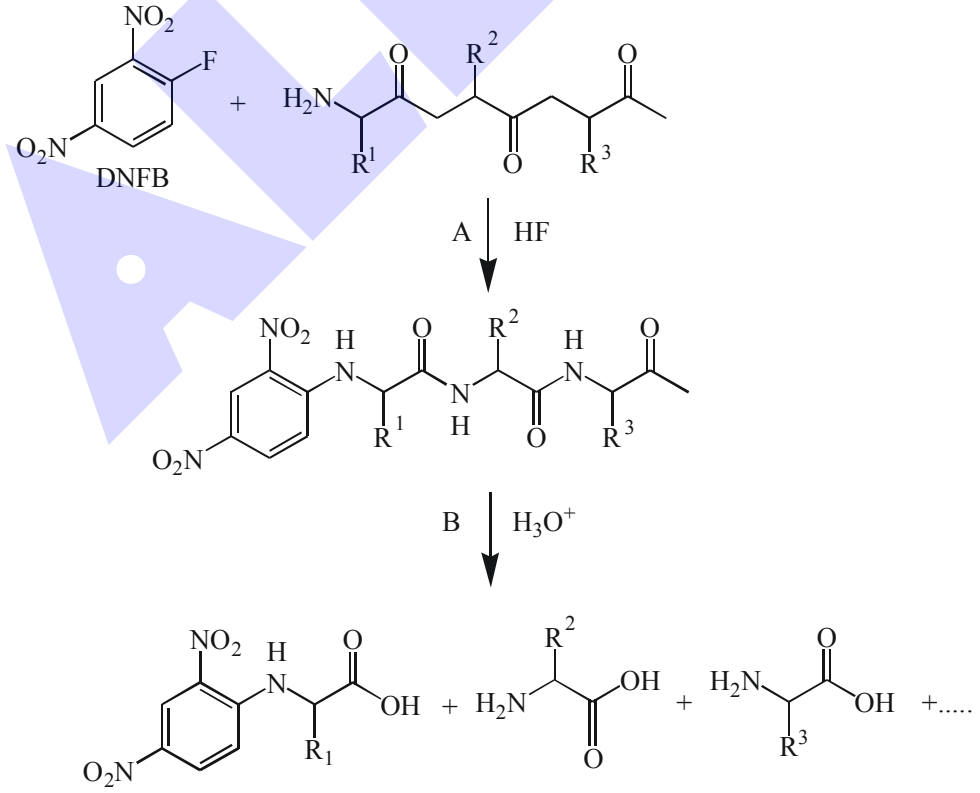
reddish-brown colored complex

Sanger's Method of sequencing of Amino acids of polypeptide chain

It is N-terminal amino acid analysis

Sanger's reagent (1-fluoro-2,4-dinitrobenzene)

1. React the peptide with a reagent that will selectively label the N-terminal amino acid.
2. Hydrolyse the protein.
3. Determine the amino acid by chromatography and comparison with standards.



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CARBOHYDRATES

Carbohydrates (hydrates of carbon) are naturally occuring compounds having general fomula $C_x(H_2O)_y$, which are constantly produced in nature & participate in many important bio-chemical reactions.

Ex. - Glucose $C_6H_{12}O_6$ $C_6(H_2O)_6$
Fructose $C_6H_{12}O_6$ $C_6(H_2O)_6$
Cellulose and Satrch $(C_6H_{10}O_5)_n$ $(C_6(H_2O)_5)_n$

- Sucrose (Cane suger) - $C_{12}H_{22}O_{11}$, and
- Maltose (Malt Suger) $C_{12}(H_2O)_{11}$

But some compounds which have formula according to $C_x(H_2O)_y$ are not known as carbohydrate

Ex. CH_2O Formaldehyde
$C_2(H_2O)_2$ Acetic acid
$C_3(H_2O)_3$ lactic acid

There are many compounds, which shows chemical behaviour of carbohydrate but do not confirm the general formula $C_x(H_2O)_y$ such as - $C_5H_{10}O_4$ (2–deoxyribose), $C_6H_{12}O_5$ (Rahmnose)
$C_7H_{14}O_6$ (Rahmnohexose)

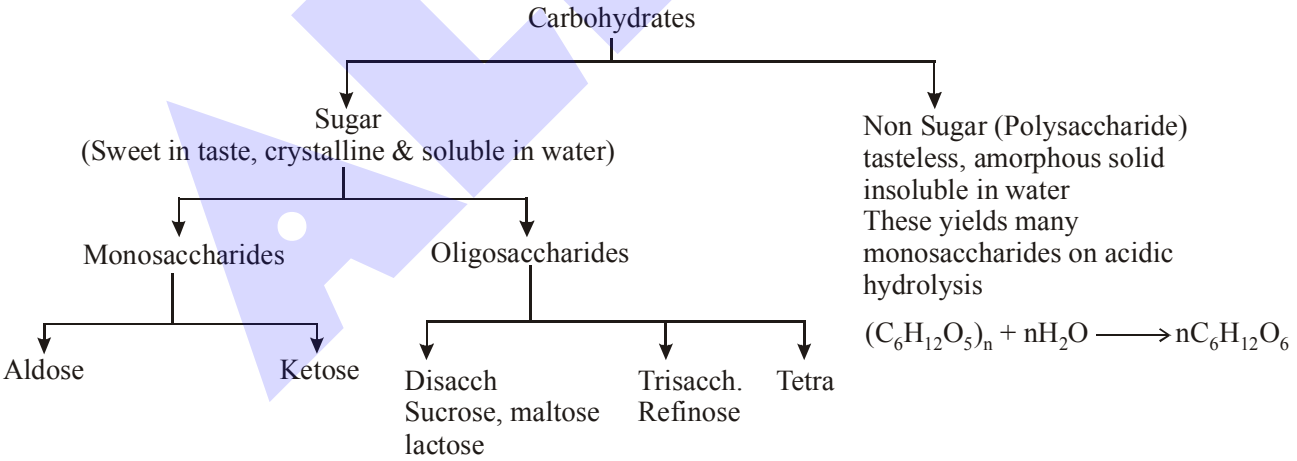
Modern Concept : Carbohydrates are polyhydroxy aldehyde or ketone

or

Substances which yield these (polyhydroxy aldehyde or ketone) on hydrolysis

- Carbohydrates $\xrightarrow{H_2O/H^+}$ Polyhydroxy aldehyde or ketone
- Carbohydrates are also known as Saccharides.
- In plants carbohydrates are synthesised by photosynthesis

Classification of carbohydrates :



Monosaccharides : (simple sugars)

These are the sugars which cannot be hydrolysed into smaller molecules.

General formula is $C_nH_{2n}O_n$

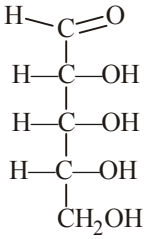
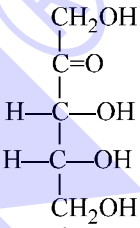
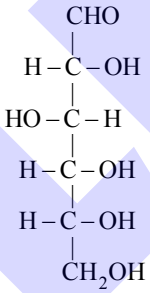
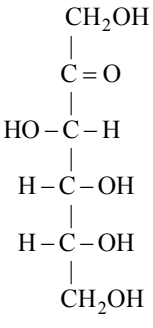
Ex. - Glucose, Fructose, Ribose, Deoxynbose

- If aldehyde group is present in monosaccharide, then it is known as aldose.
- If ketone group is present in monosaccharide , then it is known as ketose.

On the basis of C-atom monosaccharides can be farther classified as, Trioses, Tetrases, Pentoses, Hexoses.

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Table

		Aldoses	Ketoses
3C	Tropose or Triose	Aldotriose	Ketotriose
4C	Tetrose	Aldotetrose	Ketotetrose
5C	Pentose	Aldopentose	Ketopentose
5C	Including –CHO	Aldopentose (Ribose) 	
5C	Including $-\overset{\overset{O}{\|\|}}{C}-$	Ketopentose	 (Ributose)
6C	Hexose	Aldohexose	Ketohexose
6C	Including –CHO	Aldohexose (Glucose)  D – glucose	
6C	Including $-\overset{\overset{O}{\|\|}}{C}-$	Ketohexose (Fructose)	 D – fructose

Oligosaccharides :-

These are the sugars which yeilds 2–10 monosaccharides units on hydrolysis. such as.

These are of following types

(a) *Disaccharides* :- Gives two monosaccharide unit on hydrolysis (may or may not be same).

Ex. - Sucrose, Maltose, Lactose

(b) *Trisaccharides* :- Gives Three monosaccharide unit on hydrolysis.

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Polysaccharides :- These are the non sugars which yeild a large no of monosaccharide units on hydrolysis General formula - $(C_6H_{10}O_5)_n$. Ex.- Starch, Cellulose, Glycogen

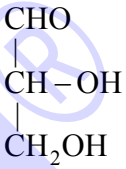
Note :- A group of polysaccharides which are not so widely used in nature is pentosans $(C_5H_8O_4)_n$

Monosaccharides, General formula $C_x(H_2O)_y$ x = 3 – 8. Nomenclature of monosaccharides are given according to the no. of carbons present in them.

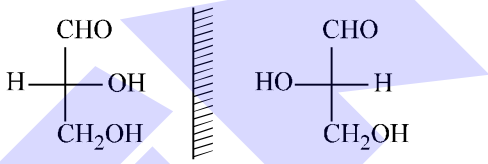
Stereochemistry of carbohydrates :

D & L-Sugars : The series of aldoses or ketoses in which the configuration of the penultimate C-atom (C-next to CH_2–OH group) is described as D-sugars if –OH is towards RHS & L-sugars if it is towards LHS.

Smallest carbohydrate
* Aldotriose
* Glyceraldehyde



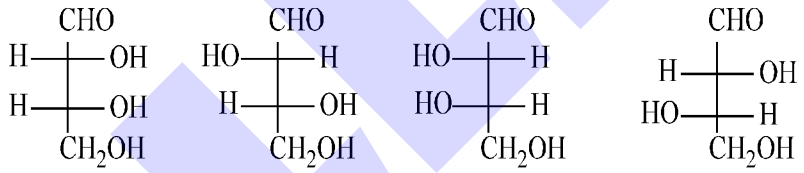
Fischer projection



D-Glyceraldehyde (+) L-Glyceraldehyde (–)

Classification of Aldotetrose :
(i) Erythrose
(ii) Threoese

C-4



D-Erythrose D-Threose L-Erythrose L-Threose

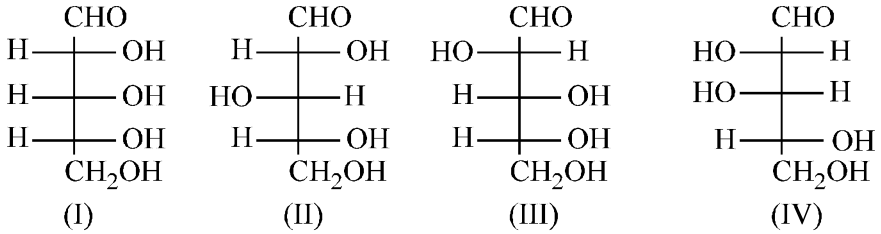
C-5



D-Aldopentose L-Aldopentose

No. of C* = 3 (in Aldopentose)
No. of optical isomers $2^3 = 8$
No. of D Sugars 4
No. of L Sugars 4

D-Aldopentose :



(I) (II) (III) (IV)

All Isomeric D-sugars are diastereomers.

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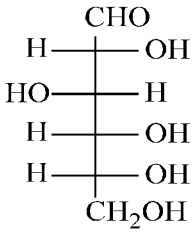
Aldohexose :

No. of C* = 4

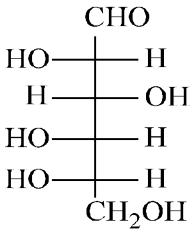
No. of stereoisomers = $2^4 = 16$

No. of D-sugars = 8

No. of L-sugars = 8

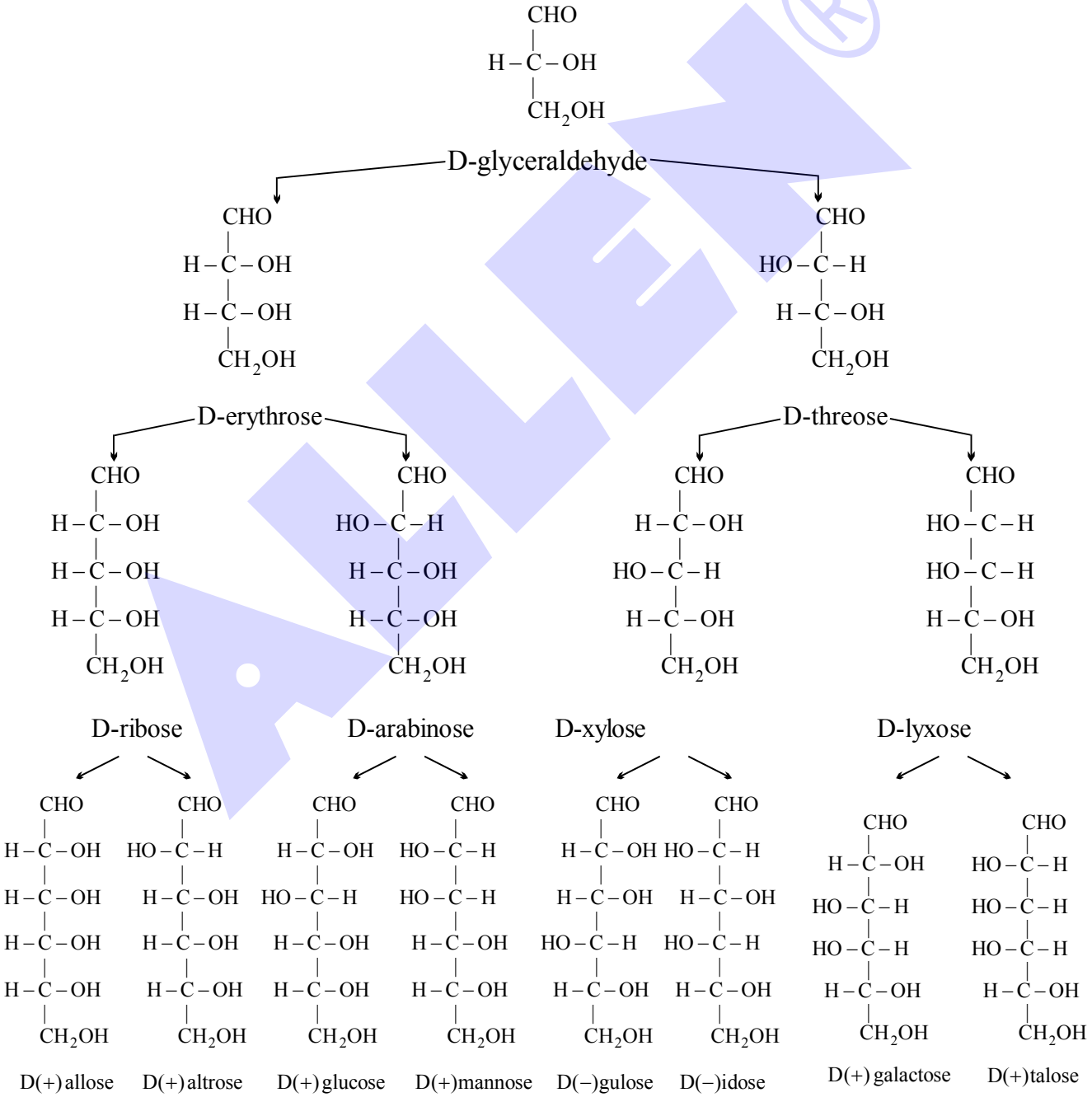


D-Glucose



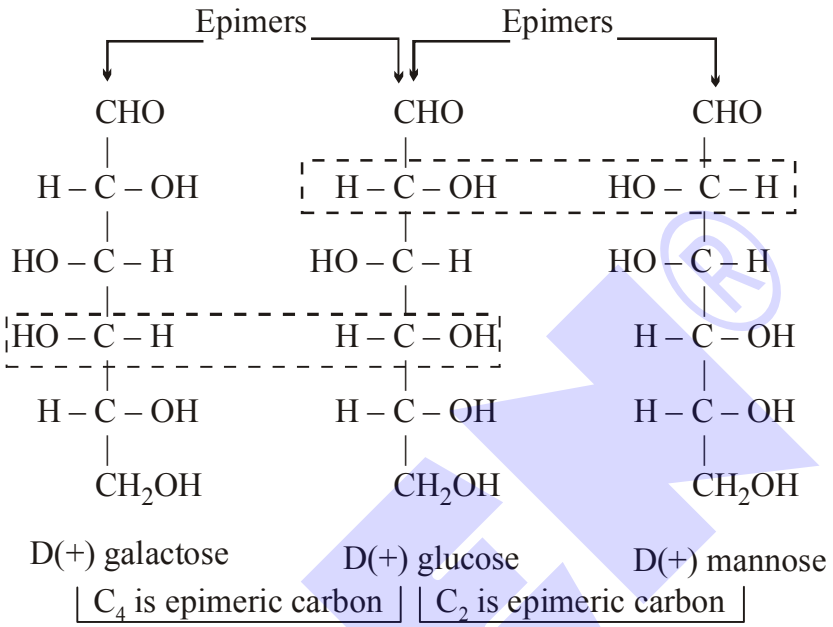
L-Glucose

The D–family aldoses



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Epimers: A pair of diastereomers that differ only in the configuration about of a single carbon atom are said to be epimers. D(+)-glucose is epimeric with D(+)- mannose and D(+)-galactose as shown below.



Example with C_2 epimeric carbon is

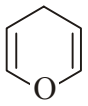


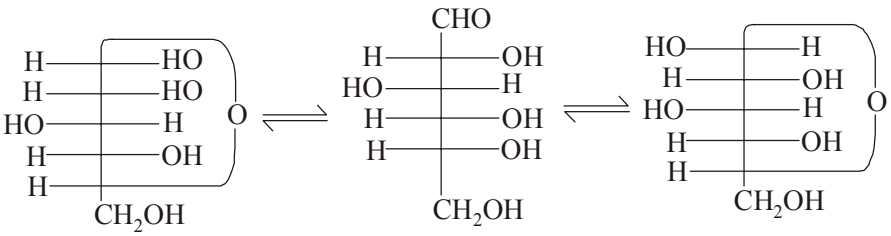
Anomers: Anomers are the stereoisomers which differs at a single chiral centre out of many & are ring chain tautomer of the same open chain compound.

The two sugars that differs in configuration only on the carbon that was the carbonyl carbon in the open chain form is called as anomers α glucose and β glucose are known as anomers their equilibrium mixture contains 36 % α–D–glucose , 63.8 % β-D-glucose and 0.2 % open chain form.

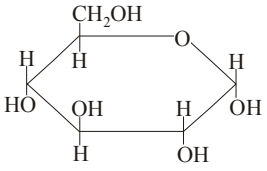
C_1 Carbon is known as anomeric carbon.

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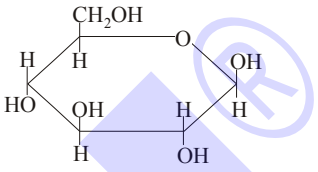
Haworth suggested to write α glucose and β glucose in pyran structure 



α–D–glucose (sp rotation 112°) | D–glucose (open structure) | β–D–glucose (sp rotation 19°)



Haworth formula α–D–glucopyranose

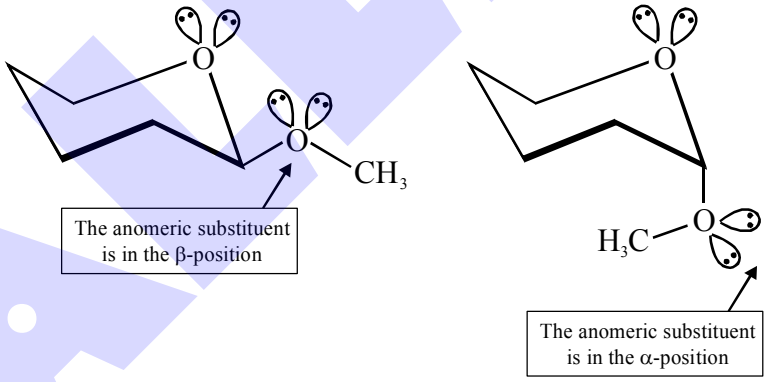


Haworth formula β–D–glucopyranose

Anomers are epimers but epimers may not be anomers.

Anomeric effect :

β-D-glucose is more stable then α-D glucose because there is more room for a substituent in the equatorial position. However when glucose reacts with an alcohol to form a glucoside, the major product is the α-glucoside. The preference for the axial position by certain substituents bonded to the anomeric carbon is called anomeric effect.



What is responsible for the anomeric effect ? One clue is that all the substituents that prefer the axial position have lone pair electrons on the atom bonded to the ring. The lone pair electrons of the anomeric substituent have repulsive interaction with the lone pair electron of the ring oxygen if the anomeric substituent is the β-position , but not if it the α-position.

Apparently attractive interaction of the hydrogen of the anomeric OH group of D-glucose with the lone pair electron of the ring oxygen decreases the importance of the anomeric effect making β-D glucose more stable than α-D glucose. However, when the hydrogen is replaced by an alkyl group, the anomeric effect decreases the stability of the β-position so, α-glycosides are more stable than β-glycosides.

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Cyclic structures of monosaccharides

Many five membered and six membered monosaccharides occur in cyclic form. Cyclic structures of monosaccharides are established by many experiments. The cyclic structure is due to intramolecular hemiacetal formation between aldo / keto group and OH of any one carbon. The ring formed are generally six membered (pyranose) or five membered (furanose). Each cyclization results in creation of a new asymmetric centre apart from the existing ones. The isomers resulting from cyclizations are called anomers. example, when D-glucose (open structure) cyclise, it gives α-D-glucose andβ-D-glucose.

Haworth projection :-

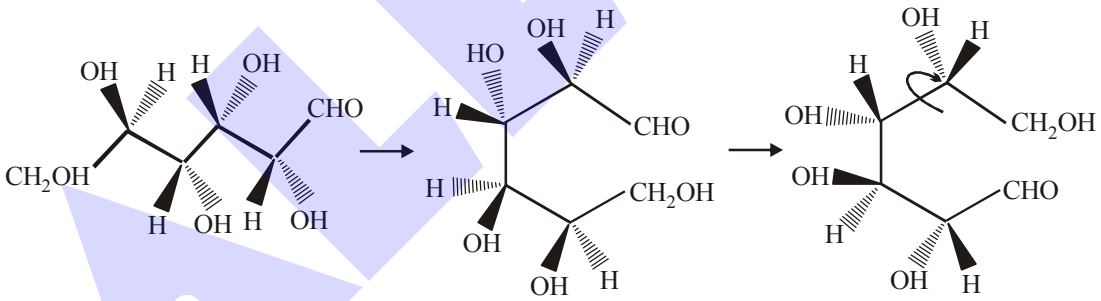
Many of monosaccharides form cyclic structures. The actual structure is almost planer and be represented by Haworth projection, which is a way of depicting three - dimensional cyclic structure.

Rule -1 :- In a Haworth projection draw a fisher projection in which ring oxygen is in a down position.

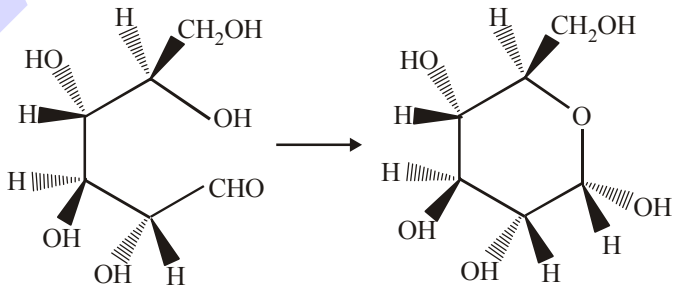
Rule -2 :- Imagine that carbon chain of fisher projection is folded around a barrel or drum, which provide a ring lies in a plane ⊥ to the page.

Rule -3 :- Now plane of ring is turned 90° so that anomeric carbon is on the right and the ring oxygen is in the rear. Obtained projection is a Haworth projection.

Example : (D-glucose)



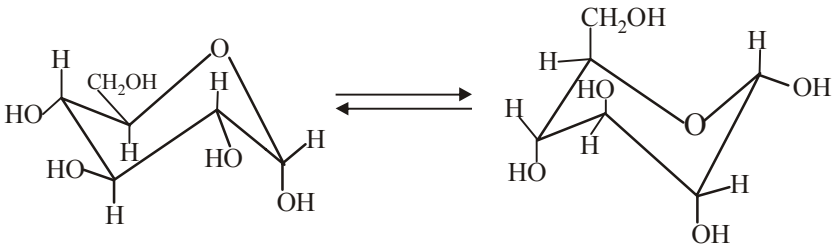
Projection :



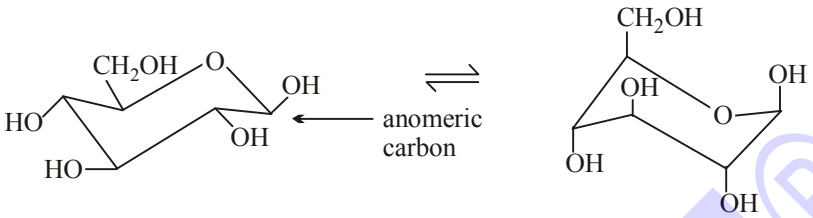
Hawarth projection

⇒ Chair conformation of D–glucose

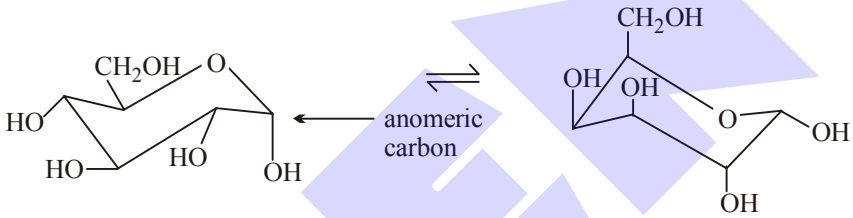
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Chair forms of (conformation) α and β D-Glucose :

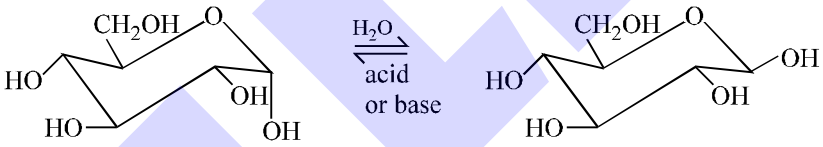


β-D-Glucose (most stable glucose form) all groups are equatorial.



α-D-Glucose –OH group at anomeric carbon is axial.

Mutarotation



α-anomer β-anomer

Specific rotation of α glucose + 112° Specific rotation of β glucose + 19°

Equilibrium mixture $[\alpha]_D = 52.5$ degree mL g^{-1} dm^{-1}

Fresh α-glucose ⟶ 52.5 ⟵ **Fresh β-glucose**

112° 36 % α glucose 19°

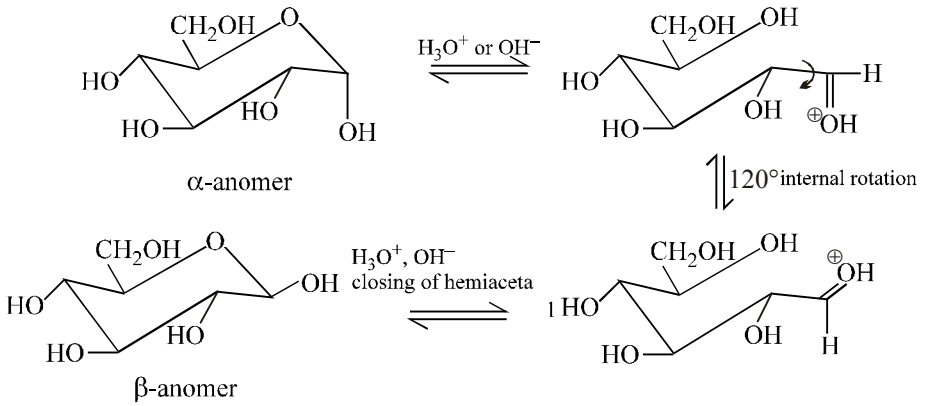
63.8 % β glucose

When pure α-D glucose is dissolved in water its specific rotation is found to be +112° with time, however the specific rotation of the solution decreases ultimately reaches stable value of +52.5°. When β D-glucose is dissolved in water, it has a specific rotation of 19°. The specific rotation of this solution increases with time also to + 52.5 °.

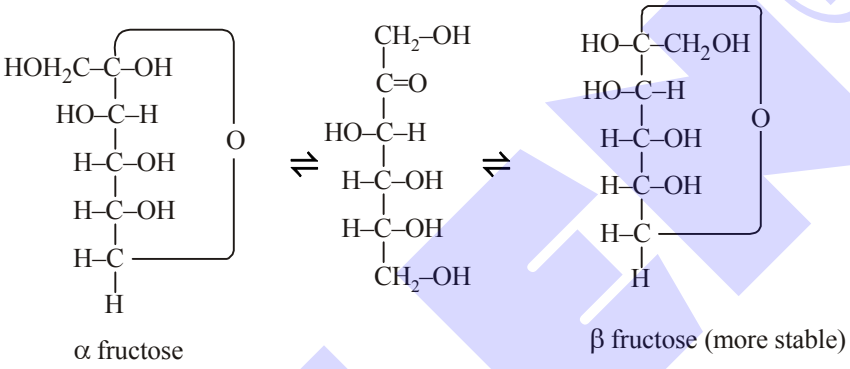
This change of optical rotation with time is called mutarotation. It is caused by the conversion of α and β glucopyranose anomers into an equilibrium mixture of both. Mutarotation is catalyzed by both acid and base, but also occurs is even in pure water. Mutarotation is characteristic of the cyclic hemiacetal form of glucose.

Mutarotation occurs first by opening of the pyranose ring to the free aldehyde form.

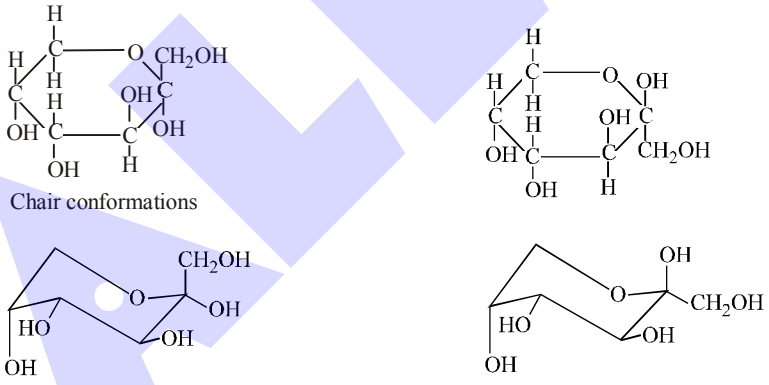
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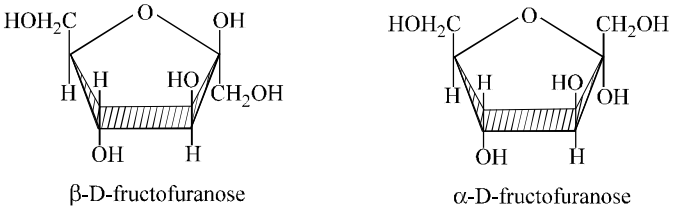
Structure of fructose



Ring structure of fructose C, Pyranose structures 6 membered ring, $C_2 - C_6$ linkage

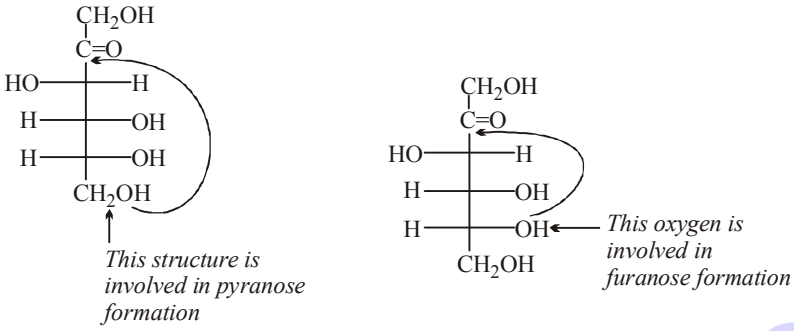


Furanose structure (5 membered ring)

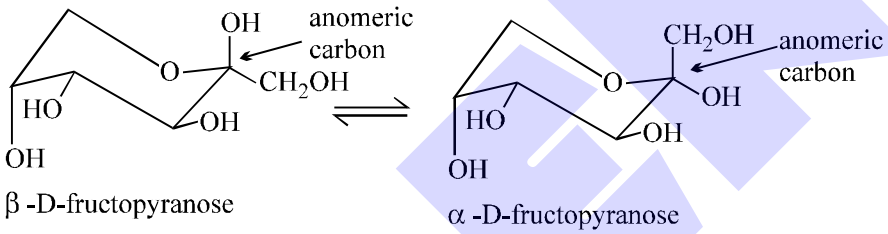


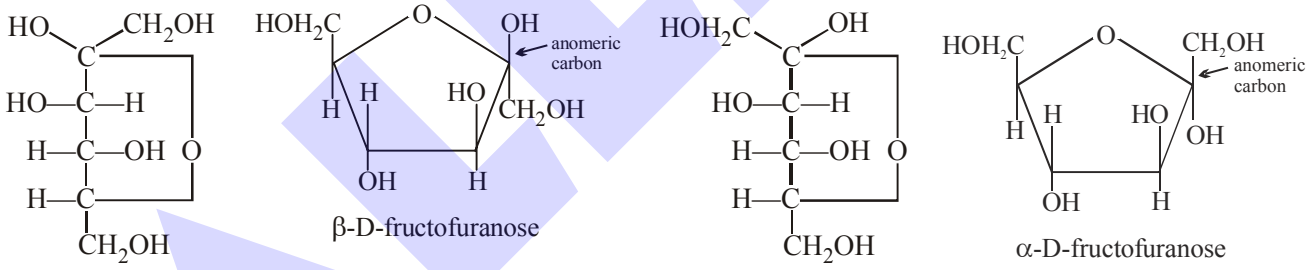
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Mutarotation: Fructose undergo complex mutarotation. The structure of the cyclic hemiacetal form of d-fructose can be derived from it's carbonyl (Ketone) form using the methods described as follows.



It happens that the crystalline form of D-fructose is β-D-Fructopyranose. When crystals of this form are dissolved in water, it equilibrates to both pyranose & furanose forms.



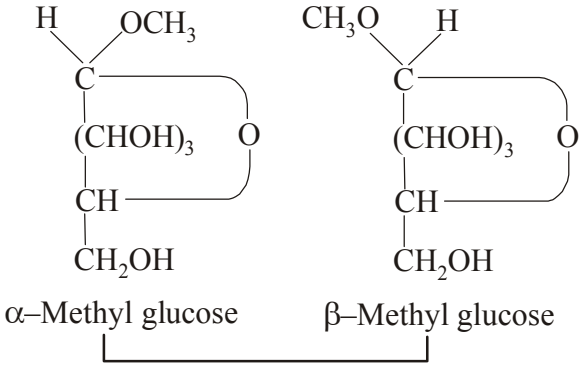


* All monosaccharides are reducing sugars and they show mutarotation.
* Starch, cellulose are Polymers of Glucose
* Lactose and sucrose are disaccharides
* Sucrose is a non reducing sugar, gives negative test for Benedict and tollen's reagent, they do not form osazone and do not show mutarotation.
* Acetals of carbohydrates are called as GLYCOSIDE

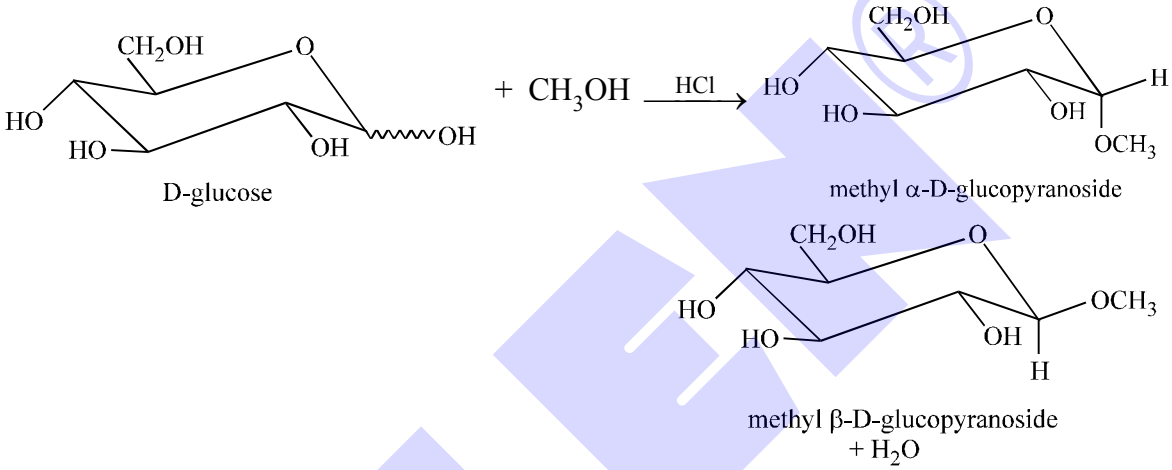
Formation of Glycosides

Glucose reacts with methyl alcohol in presence of dry HCl to form α and β-methyl glycoside of glucose. The reaction takes place only on OH of hemi-acetylic carbon. Other hydroxyl groups are unreactive.

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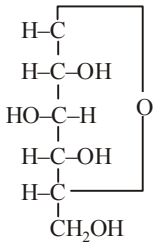


To methylate all the OH groups, methylating agent used is dimethyl sulphate.



Such compounds are called glucoside (cyclic acetals). They are special type of acetals in which one of the oxygen of the acetal linkage is the ring oxygen of the pyranose or furanose.

Ring structure of glucose :



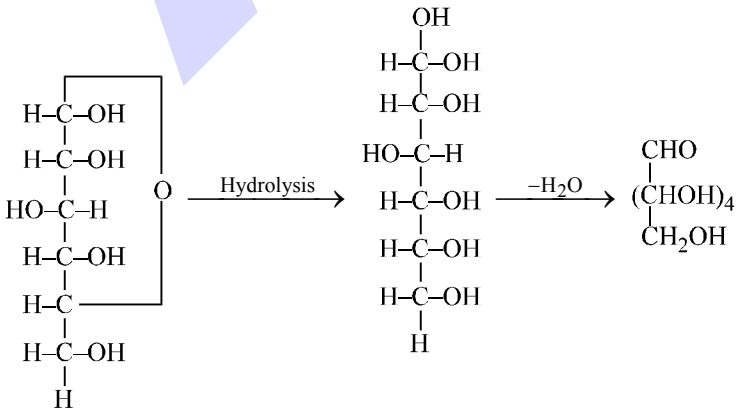
(i) Glucose does not give pink colour with schief reagent.

(ii) Does not form adduct with $NaHSO_3$, NH_3

(iii) Glucose exist in two isomeric form

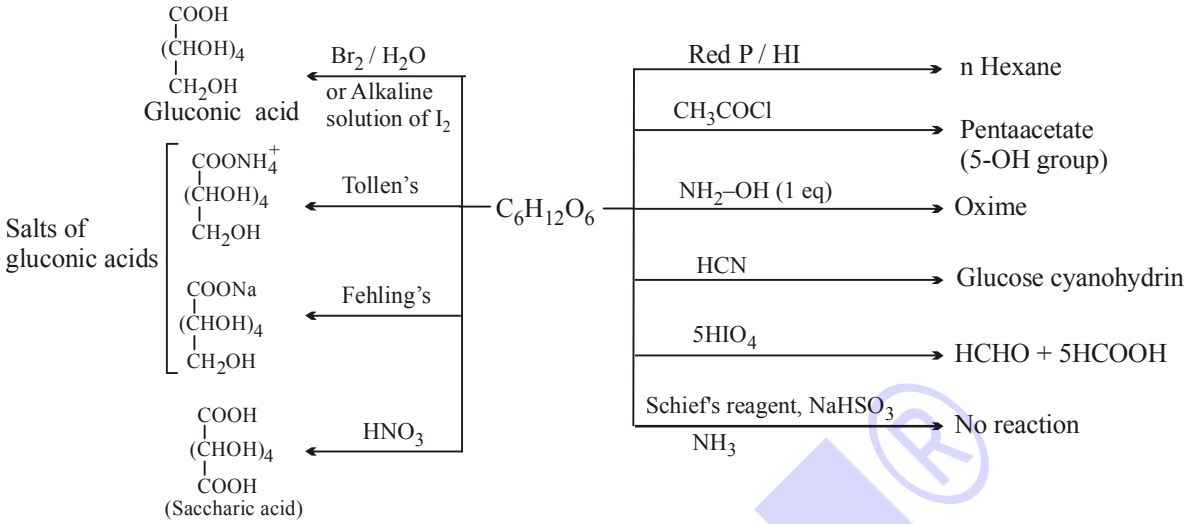
(iv) It show mutarotation

Since there is no free aldehyde group, so it does not react with weak reagent (NH_3, $NaHSO_3$) but strong reagent (HCN, NH_2OH, $C_6H_5NH-NH_2$) break up ring



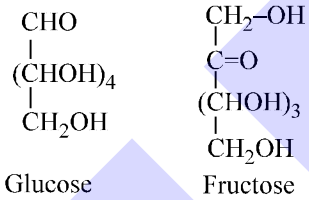
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\05-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\13-Biomole\01_Th & Ex.p65

REACTIONS OF GLUCOSE

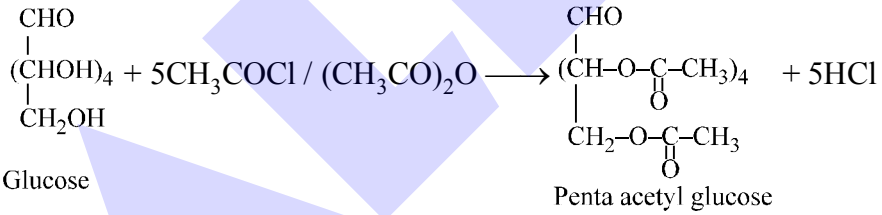


These reactions indicate that glucose has 6-C straight chain with one –CHO group & 5-OH group.

General reactions of monosaccharies



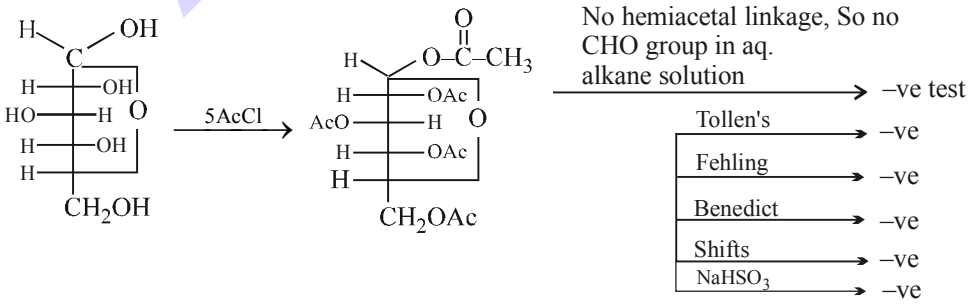
1. Acetylation :



This reaction suggests presence of 5(–OH) group.

Q. The penta acetate of glucose give –ve test with Tollen's reagent & Fehling solution, explain?

Sol.



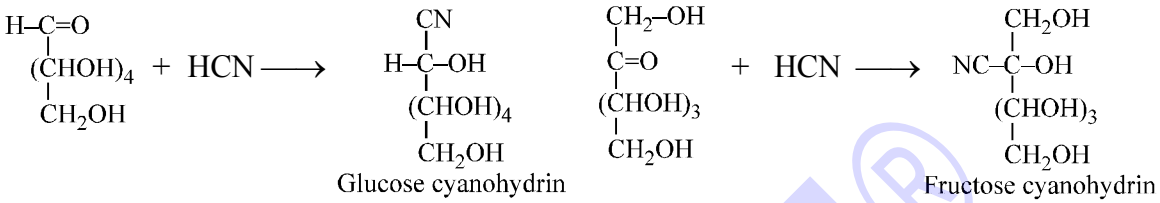
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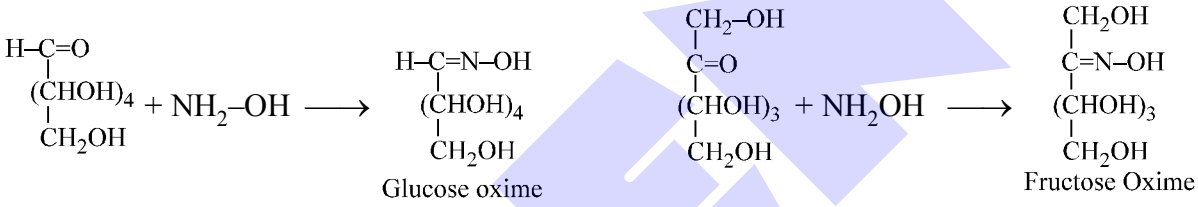
2. **Red by HI / Red P :**



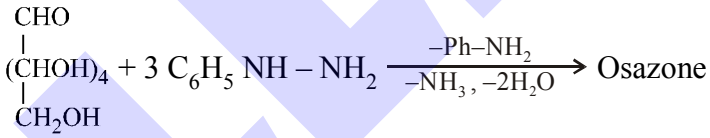
3. **Reaction with HCN:**



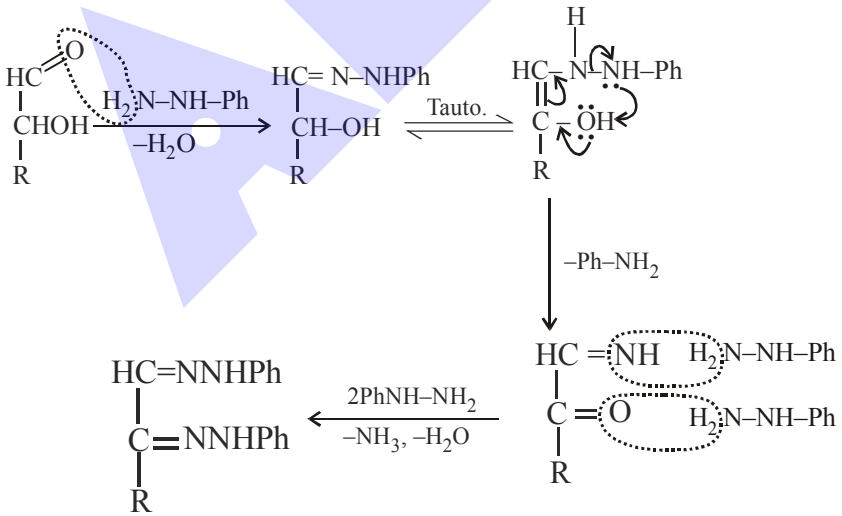
4. **Reaction with NH_2–OH (hydroxyl amine):**



5. **Reaction with phenyl hydrazine:** Both glucose and fructose give "osazone".
Reaction with glucose :

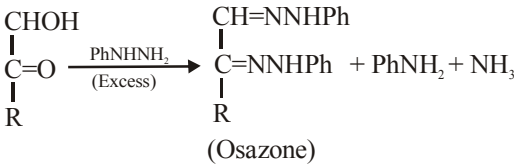


Mechanism :



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Propose Mechanism for :

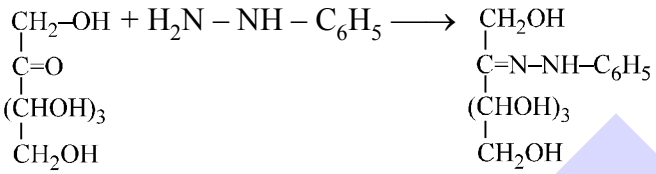


(Osazone)

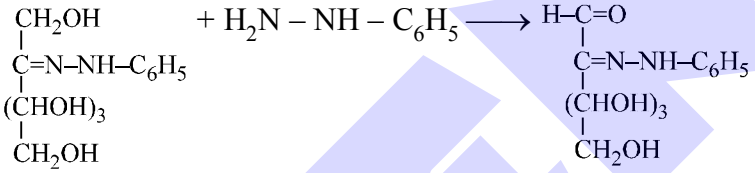
Reaction with fructose :

$$\text{Fructose} + 3\,NH_2\text{–NH–}C_6H_5 \longrightarrow \text{Osazone}$$

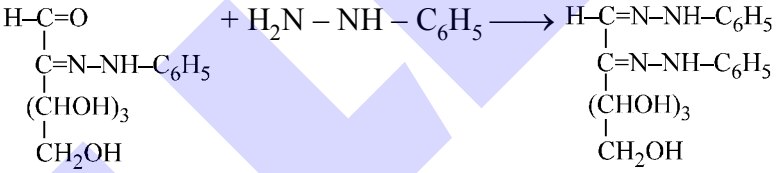
Step (i)



Step (ii)



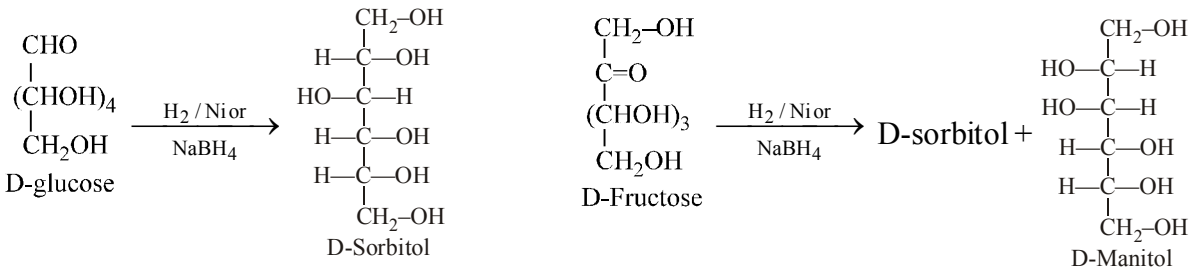
Step (iii)



Glucose , Mannose & Fructose give same osazone because structure of last four carbons in these carbohydrates is same.

Only C-1 and C-2 in glucose and fructose are involved in osazone formation addition reaction do not run through out the chain.The failure to undergo further reaction has been explained by stabilization of the osazone by chelation.

6. **Catalytic reduction:**



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7. **Oxidation:**

Glucose $\xrightarrow[\text{or } Br_2 \text{ water or Fehling solution}]{\text{Tollne's reagent}}$ $\begin{matrix}COOH\\|\\(CHOH)_4\\|\\CH_2OH\end{matrix}$ Gluconic acid

Glucose $\xrightarrow{dil.HNO_3}$ $\begin{matrix}COOH\\|\\(CHOH)_4\\|\\COOH\end{matrix}$ Saccharic acid

(Saccharic acid is also called **Glucaric acid**)

Glucose $\xrightarrow{HIO_4}$ $5HCOOH + HCHO + 5HIO_3$

Oxidation of fructose :

Fructose $\xrightarrow[\text{Fehling solution}]{\text{Tollen's, Reagent}}$ Gluconic acid

Fructose also reduces tollen's & fehling reagent because in basic medium fructose isomerises to glucose.

Fructose $\xrightarrow{Br_2 \text{water}}$ No reaction

Fructose $\xrightarrow{HNO_3}$ $\begin{matrix}CH_2OH\\|\\COOH\end{matrix}$ glyconic acid + $\begin{matrix}COOH\\|\\H-C-OH\\|\\H-C-OH\\|\\COOH\end{matrix}$ Tartaric acid (Meso) + $\begin{matrix}COOH\\|\\(CHOH)_3\\|\\COOH\end{matrix}$ Trihydroxy glutaric acid

8. **Reaction with enzyme:**

Glucose or Fructose $\xrightarrow[\text{Zymase enzyme}]{\text{Yeast}}$ $2C_2H_5OH + 2CO_2$

9. **Reaction with dil NaOH / $Ca(OH)_2$ (Base catalysed tautomerism)**
Lobry-De-Bruyn Van Ekenstein

Mannose or Glucose or fructose $\xrightarrow{\text{dil. NaOH}}$ Glucose + Fructose + Mannose

Mechanism :

$\begin{matrix}H-C=O\\|\\H-C-OH\\|\\Z\end{matrix}$ Glucose $\underset{\text{base}}{\rightleftharpoons}$ $\begin{matrix}H-C-OH\\\|\\C-OH\\|\\Z\end{matrix}$ $\rightleftharpoons$ $\begin{matrix}H-C=O\\|\\HO-C-H\\|\\Z\end{matrix}$ Mannose *(Lobry-De-Bruyn-Van Ekenstein)

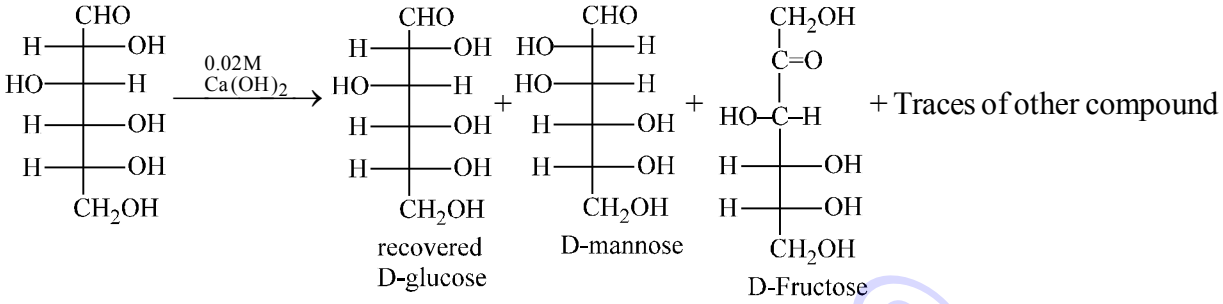
$\Downarrow\Uparrow$

$\begin{matrix}H\\|\\H-C-OH\\|\\C=O\\|\\Z\end{matrix}$ Fructose

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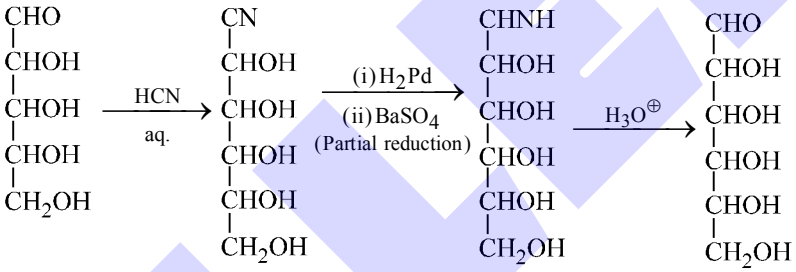
Base-catalyzed isomerisation of aldoses and Ketoses:

Although glucose in solution exists mostly in its cyclic hemiacetal forms it is also in equilibrium with a small amount of it's acyclic aldehyde form.

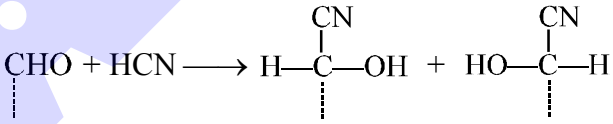


Method of ascending the sugar series: An aldose may be converted into it's next higher aldose eg. an aldopentose into an aldohexose.

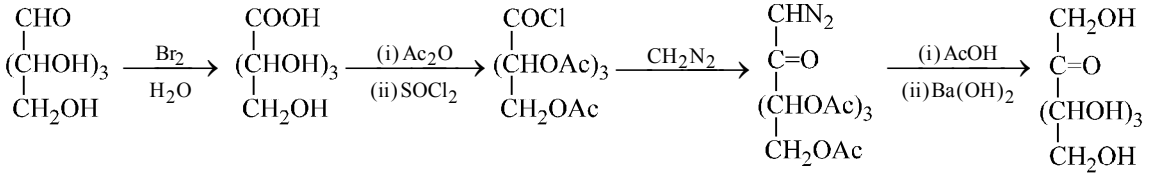
By Kiliani Fischer upgradation:



Theoretically two lactones are possible, since two cyanohydrin may be formed when hydrogen cyanide adds on to the aldopentose (a new assymetrical carbon is produced)



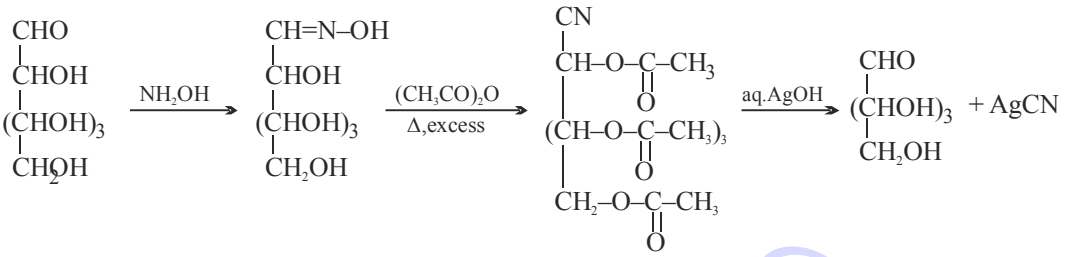
Wolfrom reaction : Wolfrom have stepped up an aldose to a ketose with one more carbon atom by a modified Arndt-Eistere reaction.



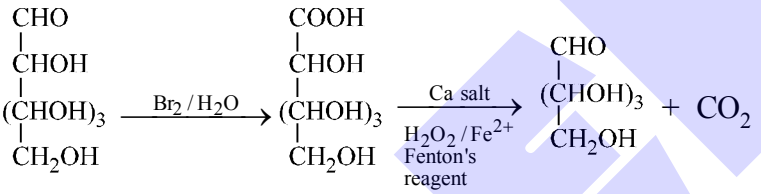
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Method of descending the sugar series :

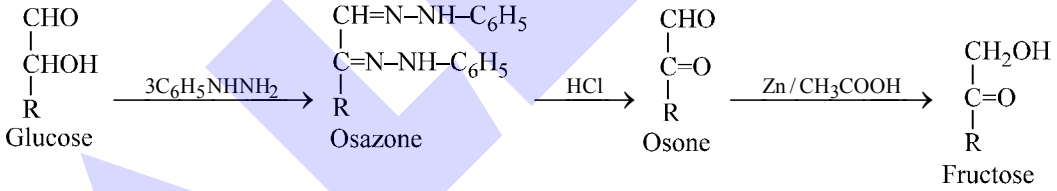
Wohl's method:



Ruff's method:

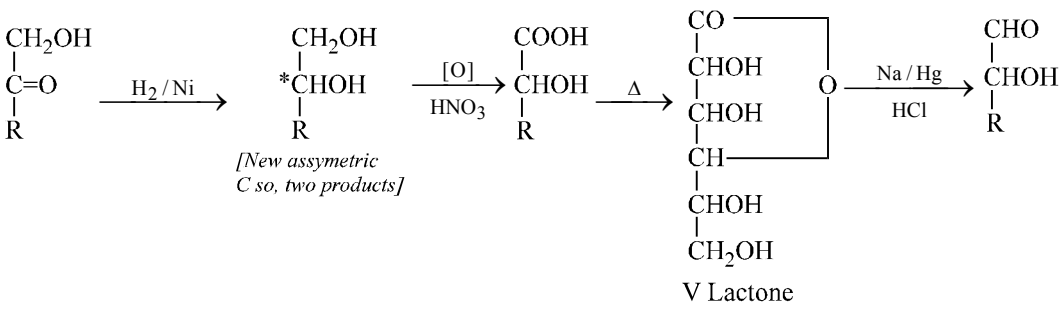


Conversion of an aldose into a ketose :



An aldehyde group is reduced more readily than a ketonic group.

Conversion of a Ketose into an aldose :



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SOME IMPORTANT CARBOHYDRATES

1. **Sucrose** ($C_{12}H_{22}O_{11}$) **:** → A dimer of α-D-Glucose & β-D-Fructose. It is white, crystalline & sweet substance soluble in water obtained from the sugar cane. When heated above its melting point. It forms a brown substance known as caramel.

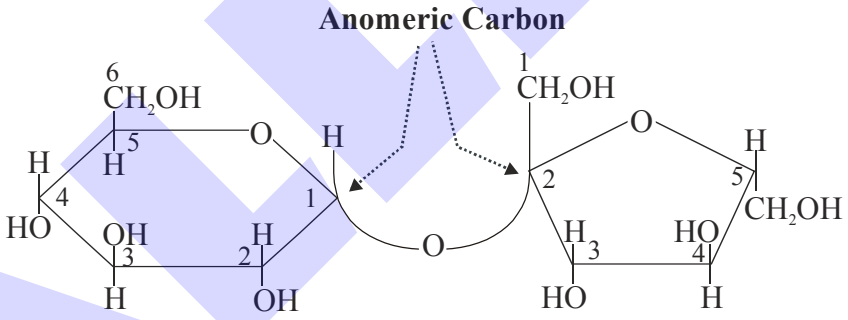
It's aqueous solution is dextrorotatory $[\propto]_D = +66.5°$

$$CH_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} \underset{\substack{C_6H_{12}O \\ \text{D-Glucose} \\ [\alpha]_D = +52.7^\circ \\ \text{Dextro Isomer}}}{\text{dextro}} + \underset{\substack{C_6H_{12}O \\ \text{D-Fructose} \\ [\alpha]_D = -92.7^\circ \\ \text{Laevo Rotatary}}}{\text{laevo}}$$

Mixture is laevorotatory

$[\alpha]_{D(\text{Mixture})} = -20^\circ$

- Thus hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to leavo (–) & such a change is known as inversion of sugar and the mixture obtained on acidic hydrolysis is known as *invert sugar*.
- The inversion of cane-sugar may also be done by the enzyme **invertase** which is found in yeast.
- Sucrose is non-reducing sugar because it has stable acetal linkage & in aq. solution it does not give free carbonyl group and so it does not reduces Tollen's & Fehling's solution.
- This indicates that neither the aldehyde group of glucose nor the ketonic group of fructose is free in sucrose.



Structure of sucrose
(α–D–glucopyranosyl–β—D–fructofuranoside)

Do not show mutarotation .

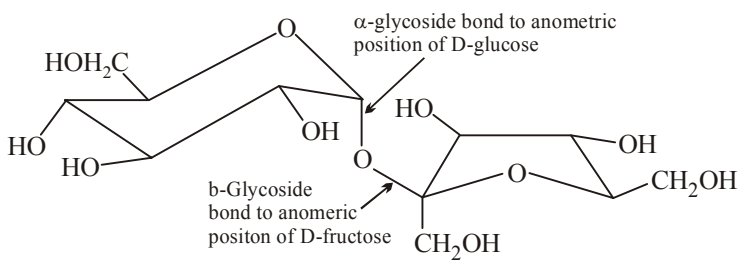


Fig. Showing Sucrose in other structure

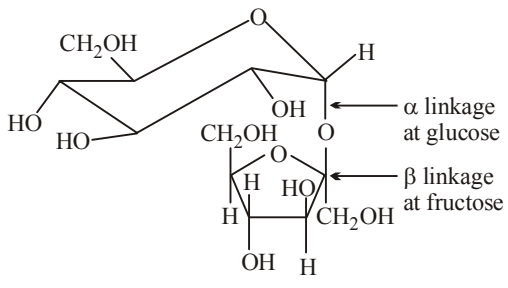


Fig. Showing Sucrose in other structure

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2. Maltose :

- It is dimer of α-D-Glucose

It is obtained by partial hydrolysis of starch by the enzyme **diastase** present in malt i.e., sprouted barely seeds.

$$2\,(C_6H_{10}O_5)_n + n\,H_2O \xrightarrow{\text{Diastase}} n\,C_{12}H_{22}O_{11}$$

Starch Maltose

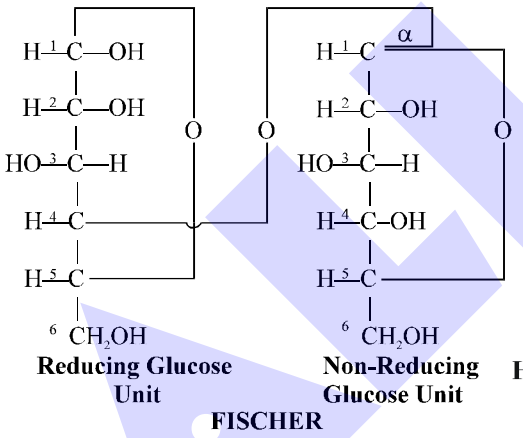
- Hydrolysis of one mole of maltose yields two moles of D-glucose.
- Maltose is a reducing sugar since it forms an osazone, undergoes mutarotation and also reduces

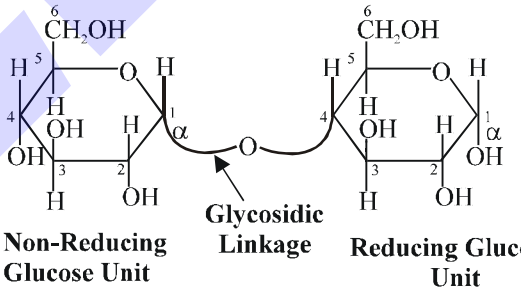
Tollen's reagents and Fehling's solutions, Methylation studies have revealed that

(i) Both glucose units are present in the pyranose form.

(ii) C_1 of one glucose unit is linked to C_4 of the other

Further since maltose is hydrolysed by the enzyme **maltase** which specifically hydrolyses α-glycosidic linkage, therefore, the non-reducing glucose unit in maltose must be present in the α-form. In other words, $C_1 - \alpha$ of non-reducing glucose unit is attached to C_4 of the reducing glucose unit as shown in the figure on next page.





HAWORTH PROJECTION FORMULAE OF α-D-MALTOSE

3. Lactose (Milk sugar) $C_{12}H_{22}O_{11}$

Lactose occurs in milk and that is why it is called milk sugar.

Lactose on hydrolysis with dilute acid or by enzyme **lactase**, yields an equimolar mixture of D-glucose and D-galactose. It is a reducing sugar it forms an osazone, undergoes mutarotation and also reduces Tollen's or Fehling's solution. Methylation studies have revealed that

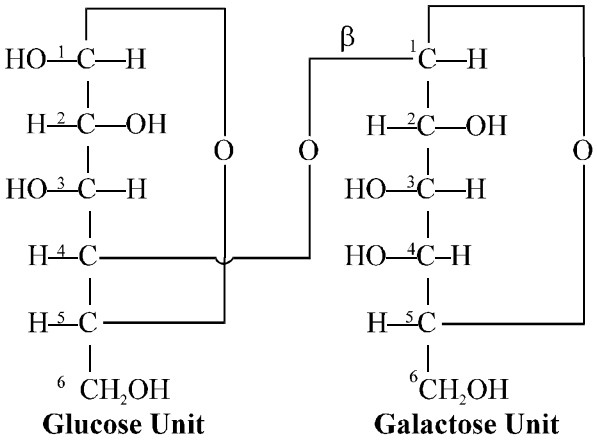
(i) both glucose and galactose are present in the pyranose form.

(ii) glucose is the reducing half while β-galactose is the non-reducing half.

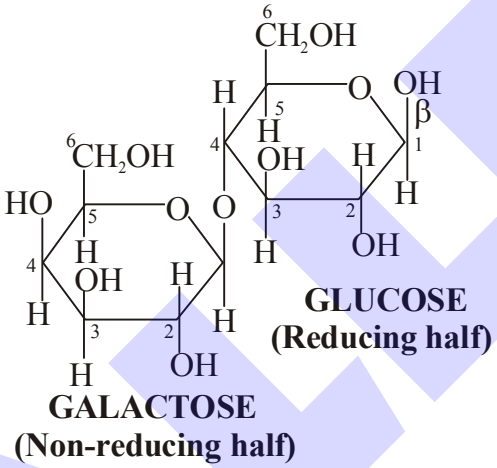
(iii) C_1 of galactose unit is connected to C_4 of glucose unit.

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Further since emulsin, as, enzyme which specifically hydrolyses β-glycosidic linkages also hydrolyses lactose, therefore, galactose must be present in the β-form. In other words, in lactose, C_1 – β of galactose is attached to C_4 of glucose as shown in figure.



FISCHER

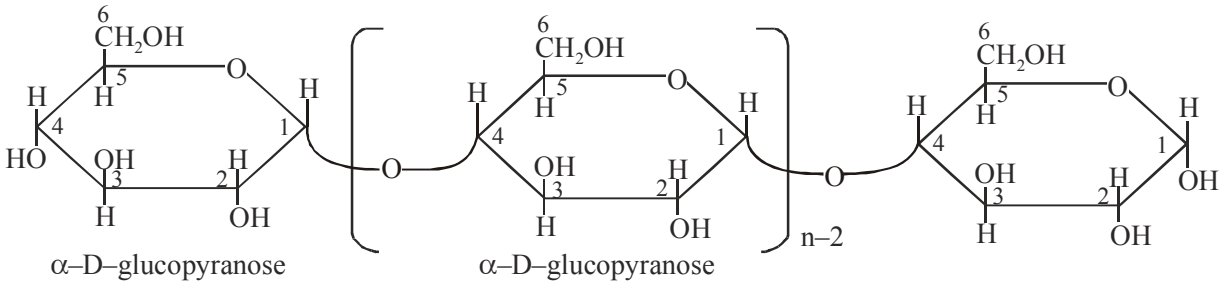


4. Starch Amylum, $(C_6H_{10}O_5)_n$

Occurrence : The value of n (100 – 3000) generally hewever it may varies from source to source. It is the chief food reserve material or storage polysaccharide of plants and is found mainly in seeds, roots tubers wheat, maize, rice, potatoes, barley, bananas and sorghum are the main sources of starch. Starch occurs in the form of granules, which vary in shape and size depending upon their plant source.

Occurs in all green plants.Starch consists of two fractions, one being known as (amylose), which gives blue colour with iodine. This blue colour is believed to be due to the formation of an inclusion complex. An aqueous solution of α–amylose slowly forms a precipitate, since α-amylose has a strong tendency to 'revert' to the insoluble state in solution. Amylopectin is insoluble in water and is stable towards both hydrolysis to maltose by the enzyme **diastase** and to D(+)-glucose by dilute acids (amylopectin gives about 50 percent of maltose).

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Structure of Starch (α–D–glucoamylose)

α–amylose consists of an unbranched chain, with a molecular weight varying between 10,000($n \approx 60$) and 10,00,000($n \approx 6,000$), The value of n depends on the source and treatment of α–amylose.

Properties : (i) Starch is a white amorphous powder sparingly soluble in water. Its aqueous solution gives a blue colour with iodine solution due to the formation of an inclusion complex. The blue pears on cooling. (ii) On hydrolysis with dilute mineral acids or enzymes, starch beaks down first to smaller molecules (n > n'), then to maltose and finally to D-glucose.

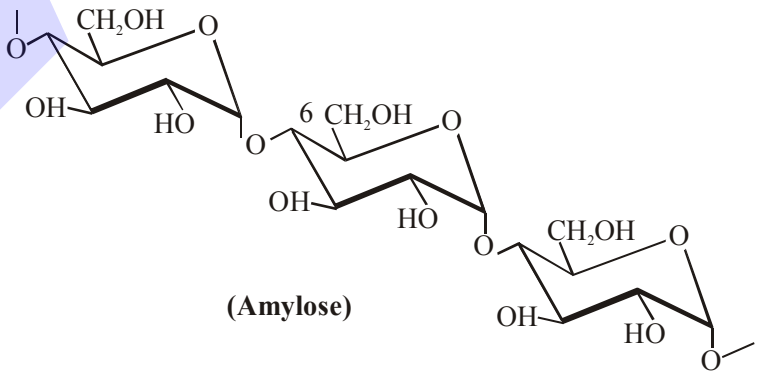
$$(C_6H_{10}O_5)n \xrightarrow{H^+/H_2O} (C_6H_{10}O_5)_{n'} \xrightarrow{H^+/H_2O} C_{12}H_{22}O_{11} \xrightarrow[\text{or Maltase}]{H^+/H_2O} C_6H_{12}O_6$$

Starch Maltose D-Glucose

(iii) Starch is a non-reducing saccharide. It neither reduces Tollen's reagent or Fehling's solution nor forms an osazone. This suggests that all hemiacetal OH groups of glucose units at C_1 are not free but are involved in glycosidic linkages.

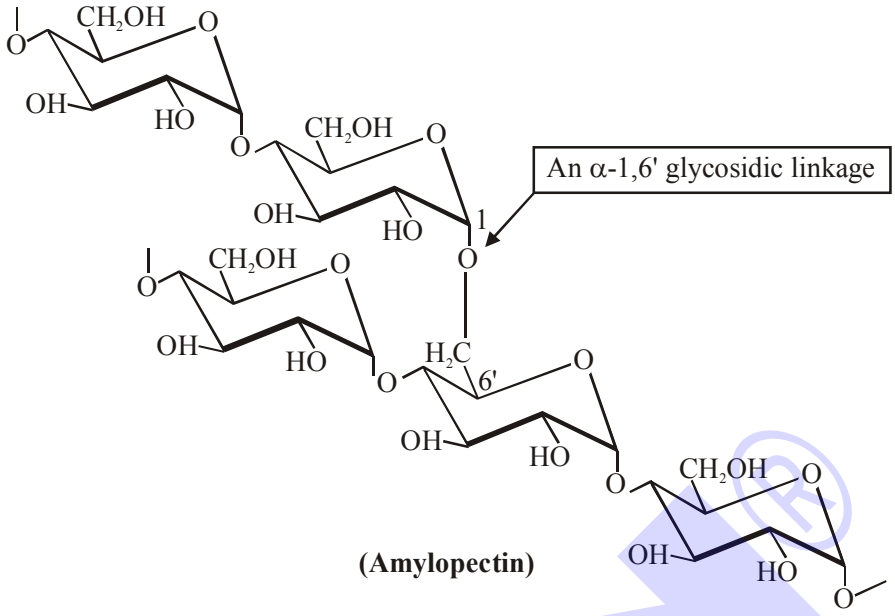
Composition : Starch is not a single compound but is a mixture of two components–a water soluble component called amylose (10-20%) and a water insoluble component called amylopectin (80-90%). Both amylose and amylopectin are polymers of α-D-glucose.

Structure of amylose : Amylose is water soluble and gives blue colour with iodine solution. It may have 100-3000 glucose units, i.e., its molecular mass can vary from 10,000 to 500,000. It is a linear polymer of α-D-glucose in which C_1 of one glucose unit is attached to C_4 of the other through α-glycosidic linkage as shown in figure.



(Amylose)

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(Amylopectin)

Pectins

Pectins are found in plant and fruit juices. Their characteristic property is the ability of their solutions to gelate, i.e. form jellies. They have a high molecular weight and are polygalacturonic acid (linked 1,4) with the carboxyl groups partially esterified with methanol.

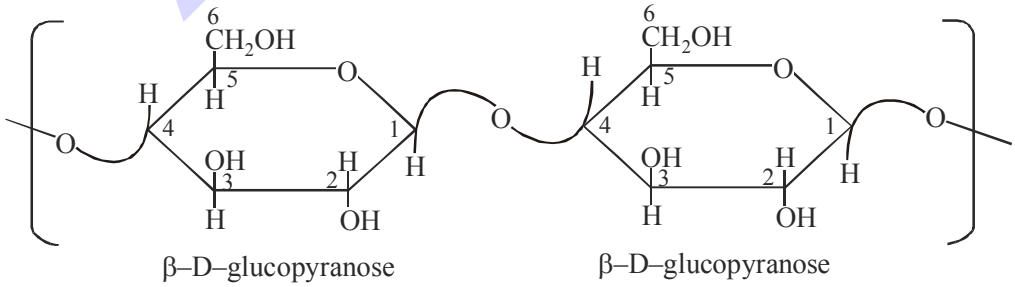
Glycogen $(C_6H_{10}O_5)_n$:

Glycogen is found in nearly all animals cells, occurring mainly in liver. It is the reserve carbohydrate of animals and so is often known as 'animal starch'. It has also been isolated from plant sources.

Glycogen is a white powder, soluble in water, the solution giving a purplish-red colour with iodine. On hydrolysis with dilute acid, glycogen gives D(+)-glucose. The molecular weight of glycogen has been given as 10,00,000 to 50,00,000 and glycogen contains highly branched chains. Glycogen has a structure similar to amylopectin, except that it has more cross-linking.

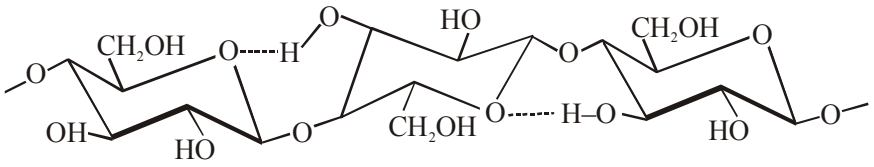
5. Cellulose:

Cellulose is colourless, solid which is insoluble in water & organic solvents. But it is soluble in ammonical cupric hydroxide (Schweizer's reagent) or in conc. HCl cellulose is a regular polymer of d-glucopyranose residues connected by β-1,4 glycosidic linkages. It is straight chain polymer.



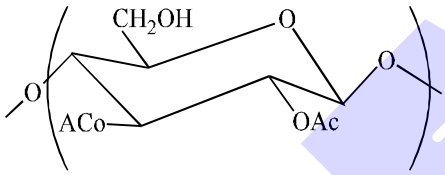
(Structure of Cellulose)

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Some points about cellulose :

1. General empirical formula ($C_6H_{10}O_5$)
2. Cellulose + H_2O $\xrightarrow{H^+}$ 96% of crystalline D-glucose
3. No. of monomer units in cellulose are 1000 – 1500 in one molecule.
4. Cellulose doesn't show mutarotation (like starch)
5. It is non reducing sugar because there is no hemiacetal linkage.
6. Acetylation, nitration & methylation of cellulose give trisubstituted cellulose which suggest that only three – OH groups are free.



Cellulose acetate

Photosynthesis

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sun light}} C_6H_{12}O_6 + 6O_2$$

* 3ATP are required to fix 1 mole of CO_2. Therefore 18 ATP are required to fix 6 mole of CO_2.

Cellular Respiration

$$C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{Enzymes}} 6CO_2 + 6H_2O + 38\ ATP$$

* 38 ATP are generated by complete oxidation of 1 mole of glucose. But net gain of ATP are 36, because 2 ATP are consumed when pyruvic acid enter in mitochondria.

* Complete oxidation takes place in two steps

Step - 1 : Glycolysis or EMP or HMP (Completes in cytoplasm)

$$C_6H_{12}O_6 \xrightarrow[2NADP \quad 2NADPH_2]{\text{Enzymes}} 2\ \underset{\text{Pyruvic acid}}{CH_3-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-OH} + 8\ ATP$$

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Step - 2 : <u>Kreb cycle (Completes in mitochondria)</u>

$$2\ CH_3-\underset{\text{Pyruvic acid}}{\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-OH} + 6O_2 \xrightarrow{\text{Enzymes}} 6CO_2 + 6H_2O + 30\ ATP$$

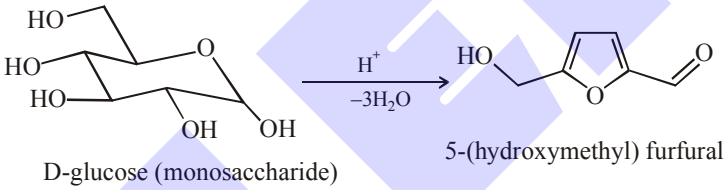
TEST OF CARBOHYDRATES

1. Molish Test	2. Barfoed's Test	3. Saliwanoff's Test.
4. Bial's Test	5. Osazone Test	6.Benedict Test
7. Fehling Test	8. Tollen's Test	9. Iodine Test

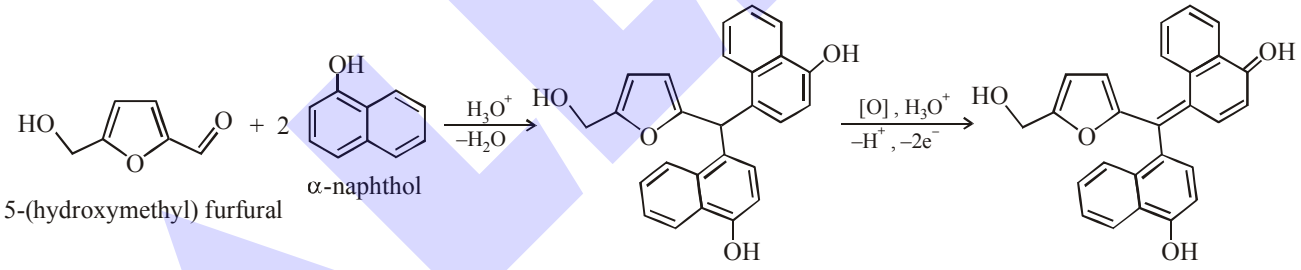
1. Molish Test

Molish test is the general test for the identification of all carbohydrates (Monosaccharides, Disaccharides and Poly saccharides) and Glycoprotein, Sulphuric acid is added to hydrolyzes the all glycosidic linkage to yield monosaccharides, which on dehydration form furfural or its derivative in presence of acid.which condened with α-naphthol to give a violet colored complex.

Oligosaccharide or poly saccharides $+ H_2SO_4 \xrightarrow{H_3O^+\text{ (Hydrolysis)}}$ Monosaccharides



D-glucose (monosaccharide) 5-(hydroxymethyl) furfural



5-(hydroxymethyl) furfural α-naphthol purple colored dye or violet colored complex

2. Barfoed's Test

This test is used to differentiate reducing monosaccharide from a disaccharide sugar

It is done in mild acidic medium.

$RCHO + 2Cu^{2+} + 2H_2O \longrightarrow RCOOH + Cu_2O\ ppt + 4H^+$

It is based on the reduction of <u>copper(II) acetate</u> to <u>copper(I) oxide</u> (Cu_2O), which forms a brick-red precipitate.

Reducing monosaccharides react with Barfoed's reagent much faster than disaccharides and produce red precipitate of copper (I) oxide within three minutes.

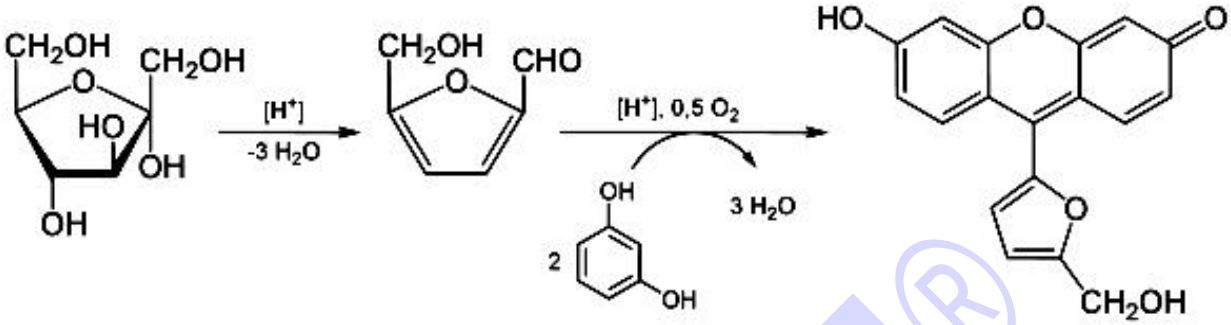
Disaccharide sugars as they are weaker reducing agents, react at a slower rate, so they do not form red precipitate even for ten minutes.

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3. Seliwanoff's Test

It is test of Ketose sugar (eg Fructose), it is used to differentiate Ketose sugar from aldose sugar.

This test relies on the principle that keto hexose are more rapidly dehydrated than aldoses to form 5-Hydroxy methyl furfural when heated in acidic medium, which on condensation with resorcinol, a cherry red (or Brown red) colored complex is formed rapidly indicating a positive test.



Fructose Resorcinol Cherry red ppt

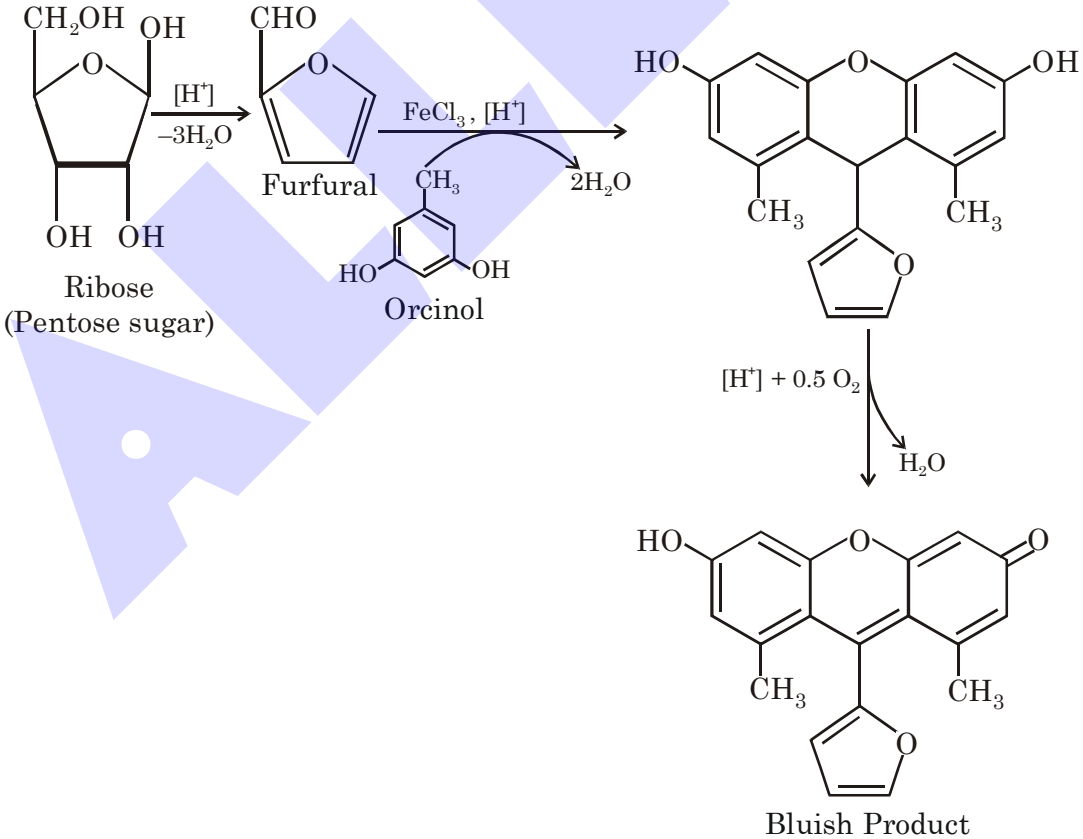
4. Bial's Test

Bial's test is positive for Pentoses

This test is used to differentiate Pentose and Hexose sugar

The test reagent dehydrates pentoses to form furfural. Furfural further reacts with Bial's reagent (a solution of orcinol, HCl and ferric chloride). orcinol and the iron ion present in the test reagent to produce a bluish product .

Figure insert



Specifically Pentose sugar gives bluish colored complex.

All other colors indicate a negative result for pentoses.

Note: hexoses generally react to form green, red, or brown products.

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5. Osazone Test

Reducing Sugars when heated with Phenyl Hydrazine,Characteristic yellow crystals of Osazone are formed with specific shape.

Glucose,Mannose & Fructose gives same Osazone crystals, like NEEDLE SHAPED.

Maltose gives Maltosazone crystals, like SUNFLOWER SHAPED.

Lactose gives lactosazone crystals,

like TIGHT BALL or POWDER PUFF SHAPED..

6. Benedict test

Carbohydrates which has Aldehyde functional group (Not Aromatic Aldehyde) or Having alpha hydroxy ketone gives positive Benedict Test

It is in mild Basic Medium.

$CuSO_4 \longrightarrow Cu^{++} + SO_4^{2-}$

$2Cu^{++}$ + Reducing Sugar $\longrightarrow Cu^{+}$

$Cu^{+} \longrightarrow Cu_2O$ ppt

Glucose (Blue Red Solution) ,,Galactose (Orange Red Solution)

Maltose (Dark Brown with Brick Red Solution),,Fructose (Dark Brown with Brick Red Solution)

Xylose (Brick Red solution)

7. Fehling's test

All reducing Carbohydrates give Positive Fehling's test with Fehling solution.Carbohydrates which has Aldehyde functional group (Not Aromatic Aldehyde) and alpha hydroxy carbonyl also gives positive Fehling's test (e.g. fuctose). While Ketones give negative fehling test. During this reaction the aldehyde group is oxidised to acid while the copper ions are reduced to red/brown precipitate of Cu_2O. This is a common test used to detect glucose in urine as positive indication of diabetes.

$CuSO_4 \longrightarrow Cu^{++} + SO_4^{2-}$

$2Cu^{++}$ + Reducing Sugar $\longrightarrow Cu_2O$ (Red / Brown ppt)

8. Tollen's test

This test is also given by reducing sugars. Carbohydrates reacts with Tollens reagent forms a silver mirror on the inner walls of the test tube. This confirms the presence of reducing sugars. Silver ions are reduced to metallic silver.

$$RCHO \xrightarrow[(ii)\,H^+]{(i)\,\left[Ag(NH_3)_2\right]OH} RCOOH + Ag \text{ (Silver mirror)}$$

Note : Sucrose , poly sachharides also others non reducing sugar do not give Benedict test , Fehling test and Tollen's Test.

9. Iodine Test

The iodine test is used to test for the presence of starch. Starch turns into an intense 'Deep blue' colour upon addition of aqueous solutions of the triodide anion.

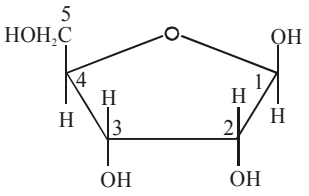
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NUCLEIC ACIDS

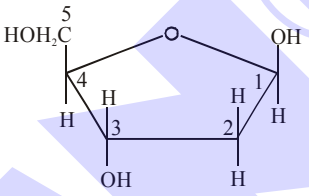
The particles in nucleus of the cell, responsible for heredity, are called chromosomes which are made up of proteins and another type of biomolecules called **nucleic acids**. These are mainly of two types, the **deoxyribonucleic acid (DNA) and ribonucleic acid (RNA)**. Since nucleic acids are long chain polymers of **nucleotides**, so they are also called polynucleotides.

Chemical Composition of Nucleic Acids

Complete hydrolysis of DNA (or RNA) yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds (called bases). In DNA molecules, the sugar moiety is β-D-2-deoxyribose whereas in RNA molecule, it is β-D-ribose.



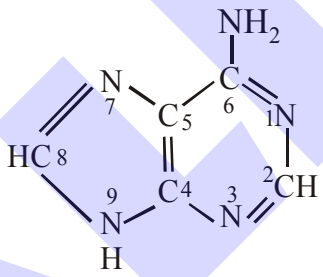
β–D–ribose



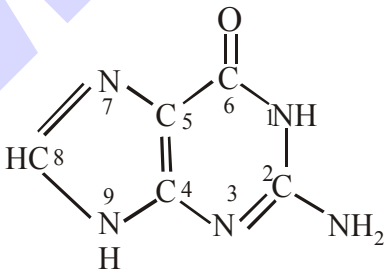
β–D–2–deoxyribose

DNA contains four bases viz. adenine (A), guanine (G), cytosine (C) and thymine (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is uracil (U).

Purines :

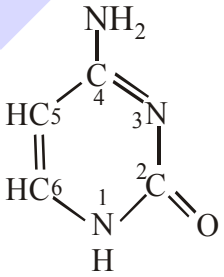


Adenine (A)

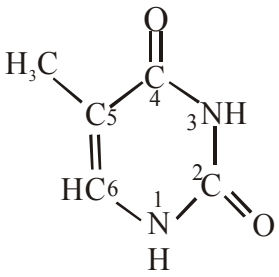


Guanine (G)

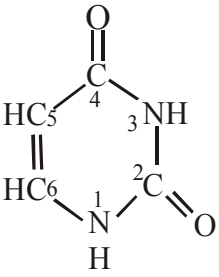
Pyrimidines :



Cytosine (C)



Thymine (T)

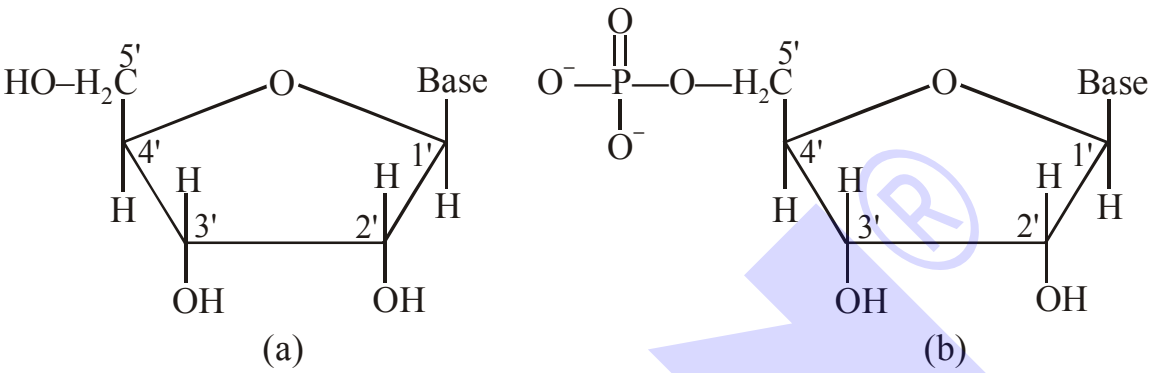


Uracil (U)

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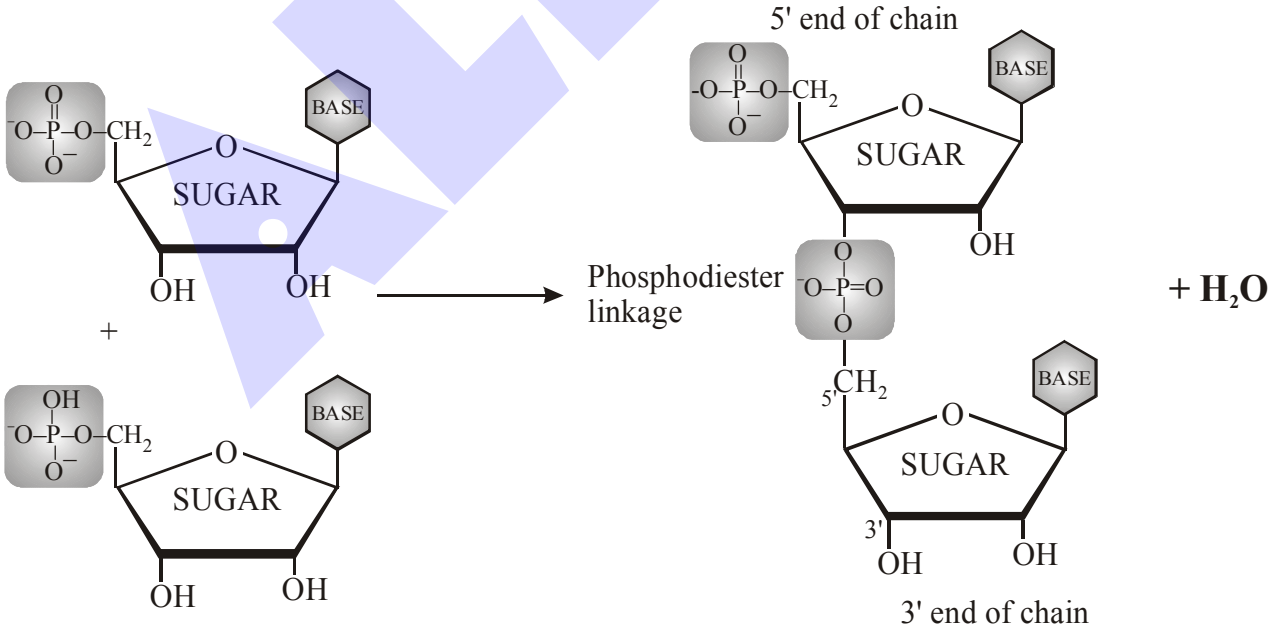
Structure of Nucleic Acids

A unit formed by the attachment of a base to 1' position of sugar is of known as **nucleoside**. In nucleosides, the sugar carbons are numbered as 1', 2', 3', etc. in order to distinguish these from the bases. When nucleoside is linked to phosphoric acid at 5'-position of sugar moiety, we get a nucleotide



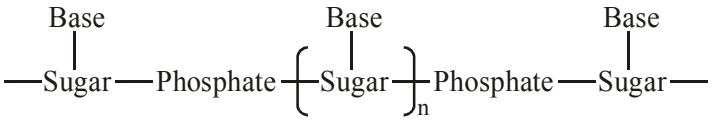
Structure of (a) a nucleoside and (b) a nucleotide

Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar. The formation of a typical dinucleotide is shown in Fig.

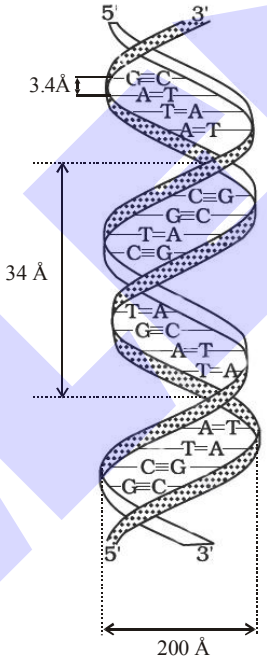


Formation of dinucleotide

A simplified version of nucleic acid chain is as shown below.



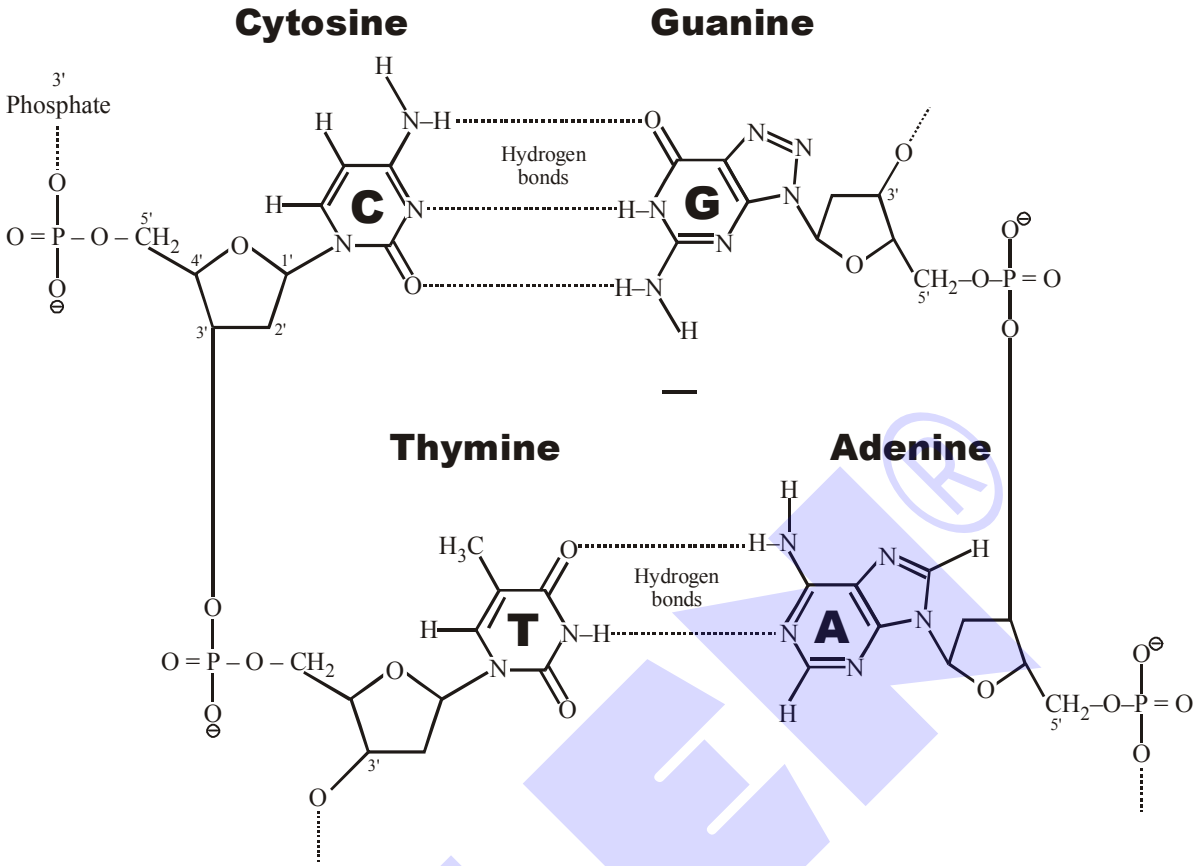
Information regarding the sequence of nucleotides in the chain of a nucleic acid is called its primary structure. Nucleic acids have a secondary structure also. James Watson and Francis Crick gav, a double strand helix structure for DNA. Two nucleic acid chains are wound about each other and held together by hydrogen bonds between pairs of bases. The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine.



In secondary structure of RNA, helices are present which are only single stranded. Sometimes they fold back on themselves to form a double helix structure. RNA molecules are of three types and they perform different functions. They are named as **messenger RNA (m-RNA), ribosomal RNA (r-RNA) and transfer RNA (t-RNA)**.

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Fig. : Double strand helix structure for DNA (Both strands are anti-parallel to each other)



A = T (Two H-bonds)

G ≡ C (Three H-bonds)

Pyrimidines

Purines

Nucleoside of RNA Nucleotide of RNA

Adenosine + Phosphate = AMP (Adenosine mono phosphate)

Guanosine + Phosphate = GMP (Guanosine mono phosphate)

Uridine + Phosphate = UMP (Uridine mono phosphate)

Cytidine + Phosphate = CMP (Cytidine mono phosphate)

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Chargaff's rule

Chargaff's rules state that DNA from any cell of all organisms should have a 1 : 1 ratio of pyrimidine and purine bases and, more specifically, that the amount of **guanine is equal to cytosine** and the amount of **adenine is equal to thymine**.

Purine = Pyrimidine

$$A + G = T + C$$

$$\frac{A+G}{T+C} = 1$$

1. **Replication :** Synthesis of DNA from DNA is called replication

$$DNA \xrightarrow[\text{Enzyme}]{\text{Replicase}} DNA$$

2. **Transcription :** Synthesis of RNA from DNA is called transcription.

$$DNA \xrightarrow[\text{Enzyme}]{\text{Transcriptase}} RNA$$

3. **Translation :** Synthesis of protein from RNA at ribosome is called translation.

$$RNA \xrightarrow{\text{Translation}} Protein$$

SOME ENZYMATIC REACTION

Material	Source	Enzymatic reaction		
Invertase	Yeast	Sucrose	→	Glucose and fructose
Zymase	Yeast	Glucose	→	Ethyl alcohol and carbon dioxide
Diastase	Malt	Starch	→	Maltose
Maltase	Yeast	Maltose	→	Glucose
Urease	Soyabean	Urea	→	Ammonia and carbon dioxide
Pepsin	Stomach	Proteins	→	Amino Acids

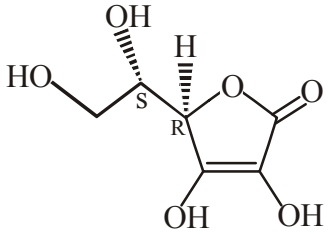
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VITAMINES

Table : Some important Vitamins, their Sources and their Deficiency Diseases

Sl. No.	*Name of Vitamins*	*Sources*	*Deficiency diseases*
1	Vitamin A	Fish liver oil, carrots, butter and milk	Xerophthamlia (hardening of cornea of eye) Night blindness
2	Vitamin B_1 (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
3	Vitamin B_2 (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis (fissuring at corners of mouth and lips), digestive disorders and burning sensation of the skin.
4	Vitamin B_6 (Pyridoxine)	Yeast, milk, egg yolk, cereals and grams	Convulsions
5	Vitamin B_{12}	Meat, fish, egg and curd	Pernicious anaemia (RBC deficient in haemoglobin)
6	Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7	Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteomalacia (soft bones and joint pain in adults)
8	Vitamin E	Vegatable oils like wheat germ oil, sunflower oil, etc.	Increased fragility of RBCs and muscular weakness
9	Vitamin K	Green leafy vegetables	Increased blood clotting time

NOTE : Vitamin-D , E , K , A are fat soluble vitamins.
Vitamin-B , C are water soluble vitamins.



Ascorbic acid
(Vitamin C)

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